



CHAPTER 11

Fourier Analysis

This chapter on Fourier analysis covers three broad areas: Fourier series in Secs. 11.1–11.4, more general orthonormal series called Sturm–Liouville expansions in Secs. 11.5 and 11.6 and Fourier integrals and transforms in Secs. 11.7–11.9.

The central starting point of Fourier analysis is **Fourier series**. They are infinite series designed to represent general periodic functions in terms of simple ones, namely, cosines and sines. This trigonometric system is *orthogonal*, allowing the computation of the coefficients of the Fourier series by use of the well-known Euler formulas, as shown in Sec. 11.1. Fourier series are very important to the engineer and physicist because they allow the solution of ODEs in connection with forced oscillations (Sec. 11.3) and the approximation of periodic functions (Sec. 11.4). Moreover, applications of Fourier analysis to PDEs are given in Chap. 12. Fourier series are, in a certain sense, more universal than the familiar Taylor series in calculus because many *discontinuous* periodic functions that come up in applications can be developed in Fourier series but do not have Taylor series expansions.

The underlying idea of the Fourier series can be extended in two important ways. We can replace the trigonometric system by other families of orthogonal functions, e.g., Bessel functions and obtain the **Sturm–Liouville expansions**. Note that related Secs. 11.5 and 11.6 used to be part of Chap. 5 but, for greater readability and logical coherence, are now part of Chap. 11. The second expansion is applying Fourier series to nonperiodic phenomena and obtaining Fourier integrals and Fourier transforms. Both extensions have important applications to solving PDEs as will be shown in Chap. 12.

In a digital age, the *discrete Fourier transform* plays an important role. Signals, such as voice or music, are sampled and analyzed for frequencies. An important algorithm, in this context, is the *fast Fourier transform*. This is discussed in Sec. 11.9.

Note that the two extensions of Fourier series are independent of each other and may be studied in the order suggested in this chapter or by studying Fourier integrals and transforms first and then Sturm–Liouville expansions.

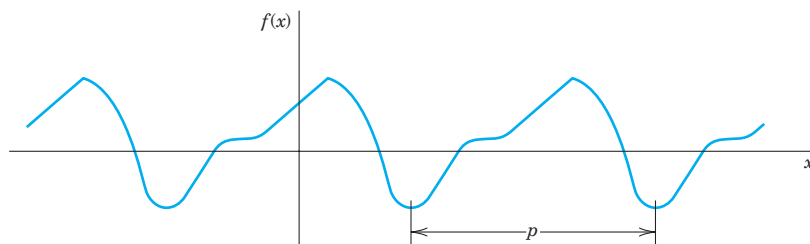
Prerequisite: Elementary integral calculus (needed for Fourier coefficients).

Sections that may be omitted in a shorter course: 11.4–11.9.

References and Answers to Problems: App. 1 Part C, App. 2.

11.1 Fourier Series

Fourier series are infinite series that represent periodic functions in terms of cosines and sines. As such, Fourier series are of greatest importance to the engineer and applied mathematician. To define Fourier series, we first need some background material. A function $f(x)$ is called a **periodic function** if $f(x)$ is defined for all real x , except

Fig. 258. Periodic function of period p

possibly at some points, and if there is some positive number p , called a **period** of $f(x)$, such that

$$(1) \quad f(x + p) = f(x) \quad \text{for all } x.$$

(The function $f(x) = \tan x$ is a periodic function that is not defined for all real x but undefined for some points (more precisely, countably many points), that is $x = \pm\pi/2, \pm3\pi/2, \dots$)

The graph of a periodic function has the characteristic that it can be obtained by periodic repetition of its graph in any interval of length p (Fig. 258).

The smallest positive period is often called the *fundamental period*. (See Probs. 2–4.)

Familiar periodic functions are the cosine, sine, tangent, and cotangent. Examples of functions that are not periodic are $x, x^2, x^3, e^x, \cosh x$, and $\ln x$, to mention just a few.

If $f(x)$ has period p , it also has the period $2p$ because (1) implies $f(x + 2p) = f([x + p] + p) = f(x + p) = f(x)$, etc.; thus for any integer $n = 1, 2, 3, \dots$,

$$(2) \quad f(x + np) = f(x) \quad \text{for all } x.$$

Furthermore if $f(x)$ and $g(x)$ have period p , then $af(x) + bg(x)$ with any constants a and b also has the period p .

Our problem in the first few sections of this chapter will be the representation of various **functions $f(x)$ of period 2π** in terms of the simple functions

$$(3) \quad 1, \quad \cos x, \quad \sin x, \quad \cos 2x, \quad \sin 2x, \dots, \quad \cos nx, \quad \sin nx, \dots$$

All these functions have the period 2π . They form the so-called **trigonometric system**. Figure 259 shows the first few of them (except for the constant 1, which is periodic with any period).

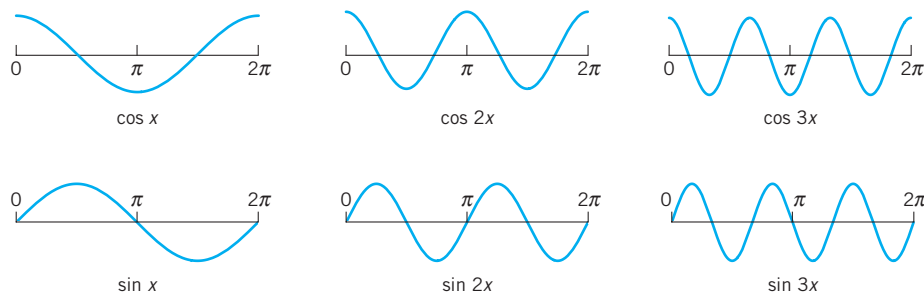


Fig. 259. Cosine and sine functions having the period 2π (the first few members of the trigonometric system (3), except for the constant 1)

The series to be obtained will be a **trigonometric series**, that is, a series of the form

$$\begin{aligned} & a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \cdots \\ (4) \quad & = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx). \end{aligned}$$

$a_0, a_1, b_1, a_2, b_2, \dots$ are constants, called the **coefficients** of the series. We see that each term has the period 2π . Hence *if the coefficients are such that the series converges, its sum will be a function of period 2π .*

Expressions such as (4) will occur frequently in Fourier analysis. To compare the expression on the right with that on the left, simply write the terms in the summation. Convergence of one side implies convergence of the other and the sums will be the same.

Now suppose that $f(x)$ is a given function of period 2π and is such that it can be **represented** by a series (4), that is, (4) converges and, moreover, has the sum $f(x)$. Then, using the equality sign, we write

$$(5) \quad f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

and call (5) the **Fourier series** of $f(x)$. We shall prove that in this case the coefficients of (5) are the so-called **Fourier coefficients** of $f(x)$, given by the **Euler formulas**

$$\begin{aligned} (0) \quad & a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx \\ (6) \quad (a) \quad & a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \quad n = 1, 2, \dots \\ (b) \quad & b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx \quad n = 1, 2, \dots \end{aligned}$$

The name “Fourier series” is sometimes also used in the exceptional case that (5) with coefficients (6) does not converge or does not have the sum $f(x)$ —this may happen but is merely of theoretical interest. (For Euler see footnote 4 in Sec. 2.5.)

A Basic Example

Before we derive the Euler formulas (6), let us consider how (5) and (6) are applied in this important basic example. Be fully alert, as the way we approach and solve this example will be the technique you will use for other functions. Note that the integration is a little bit different from what you are familiar with in calculus because of the n . Do not just routinely use your software but try to get a good understanding and make observations: How are continuous functions (cosines and sines) able to represent a given discontinuous function? How does the quality of the approximation increase if you take more and more terms of the series? Why are the approximating functions, called the

partial sums of the series, in this example always zero at 0 and π ? Why is the factor $1/n$ (obtained in the integration) important?

EXAMPLE 1 Periodic Rectangular Wave (Fig. 260)

Find the Fourier coefficients of the periodic function $f(x)$ in Fig. 260. The formula is

$$(7) \quad f(x) = \begin{cases} -k & \text{if } -\pi < x < 0 \\ k & \text{if } 0 < x < \pi \end{cases} \quad \text{and} \quad f(x + 2\pi) = f(x).$$

Functions of this kind occur as external forces acting on mechanical systems, electromotive forces in electric circuits, etc. (The value of $f(x)$ at a single point does not affect the integral; hence we can leave $f(x)$ undefined at $x = 0$ and $x = \pm\pi$.)

Solution. From (6.0) we obtain $a_0 = 0$. This can also be seen without integration, since the area under the curve of $f(x)$ between $-\pi$ and π (taken with a minus sign where $f(x)$ is negative) is zero. From (6a) we obtain the coefficients $a_1, a_2, \dots$ of the cosine terms. Since $f(x)$ is given by two expressions, the integrals from $-\pi$ to π split into two integrals:

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = \frac{1}{\pi} \left[\int_{-\pi}^0 (-k) \cos nx \, dx + \int_0^{\pi} k \cos nx \, dx \right] \\ &= \frac{1}{\pi} \left[-k \frac{\sin nx}{n} \Big|_{-\pi}^0 + k \frac{\sin nx}{n} \Big|_0^{\pi} \right] = 0 \end{aligned}$$

because $\sin nx = 0$ at $-\pi, 0$, and π for all $n = 1, 2, \dots$. We see that all these cosine coefficients are zero. That is, the Fourier series of (7) has no cosine terms, just sine terms, it is a **Fourier sine series** with coefficients $b_1, b_2, \dots$ obtained from (6b);

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = \frac{1}{\pi} \left[\int_{-\pi}^0 (-k) \sin nx \, dx + \int_0^{\pi} k \sin nx \, dx \right] \\ &= \frac{1}{\pi} \left[k \frac{\cos nx}{n} \Big|_{-\pi}^0 - k \frac{\cos nx}{n} \Big|_0^{\pi} \right]. \end{aligned}$$

Since $\cos(-\alpha) = \cos \alpha$ and $\cos 0 = 1$, this yields

$$b_n = \frac{k}{n\pi} [\cos 0 - \cos(-n\pi) - \cos n\pi + \cos 0] = \frac{2k}{n\pi} (1 - \cos n\pi).$$

Now, $\cos \pi = -1$, $\cos 2\pi = 1$, $\cos 3\pi = -1$, etc.; in general,

$$\cos n\pi = \begin{cases} -1 & \text{for odd } n, \\ 1 & \text{for even } n, \end{cases} \quad \text{and thus} \quad 1 - \cos n\pi = \begin{cases} 2 & \text{for odd } n, \\ 0 & \text{for even } n. \end{cases}$$

Hence the Fourier coefficients b_n of our function are

$$b_1 = \frac{4k}{\pi}, \quad b_2 = 0, \quad b_3 = \frac{4k}{3\pi}, \quad b_4 = 0, \quad b_5 = \frac{4k}{5\pi}, \dots$$

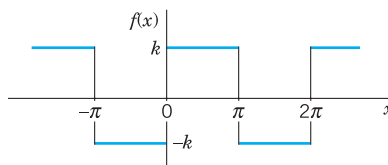


Fig. 260. Given function $f(x)$ (Periodic rectangular wave)

Since the a_n are zero, the Fourier series of $f(x)$ is

$$(8) \quad \frac{4k}{\pi} \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \cdots \right).$$

The partial sums are

$$S_1 = \frac{4k}{\pi} \sin x, \quad S_2 = \frac{4k}{\pi} \left(\sin x + \frac{1}{3} \sin 3x \right), \quad \text{etc.}$$

Their graphs in Fig. 261 seem to indicate that the series is convergent and has the sum $f(x)$, the given function. We notice that at $x = 0$ and $x = \pi$, the points of discontinuity of $f(x)$, all partial sums have the value zero, the arithmetic mean of the limits $-k$ and k of our function, at these points. This is typical.

Furthermore, assuming that $f(x)$ is the sum of the series and setting $x = \pi/2$, we have

$$f\left(\frac{\pi}{2}\right) = k = \frac{4k}{\pi} \left(1 - \frac{1}{3} + \frac{1}{5} - + \cdots \right).$$

Thus

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + - \cdots = \frac{\pi}{4}.$$

This is a famous result obtained by Leibniz in 1673 from geometric considerations. It illustrates that the values of various series with constant terms can be obtained by evaluating Fourier series at specific points. ■

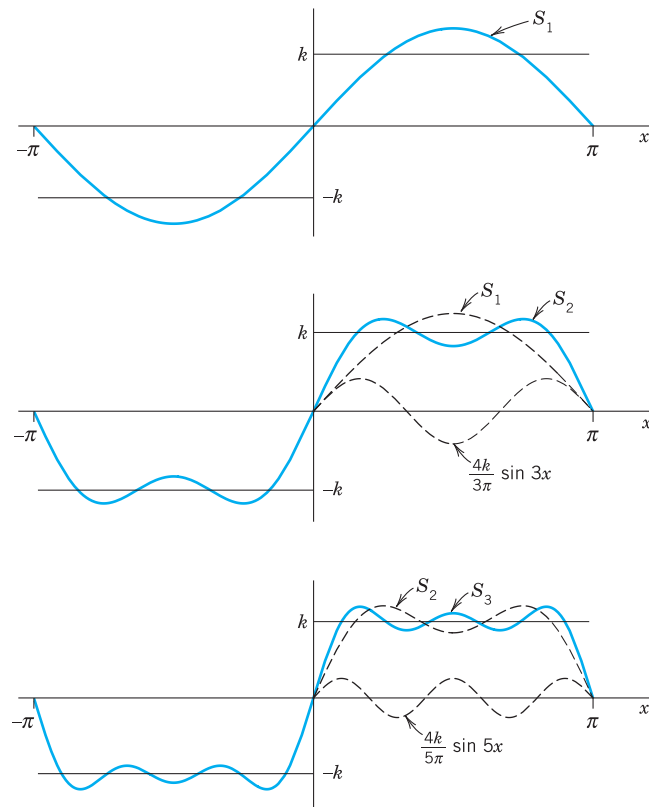


Fig. 261. First three partial sums of the corresponding Fourier series

Derivation of the Euler Formulas (6)

The key to the Euler formulas (6) is the **orthogonality** of (3), a concept of basic importance, as follows. Here we generalize the concept of inner product (Sec. 9.3) to functions.

THEOREM 1

Orthogonality of the Trigonometric System (3)

The trigonometric system (3) is orthogonal on the interval $-\pi \leq x \leq \pi$ (hence also on $0 \leq x \leq 2\pi$ or any other interval of length 2π because of periodicity); that is, the integral of the product of any two functions in (3) over that interval is 0, so that for any integers n and m ,

$$(9) \quad \begin{aligned} (a) \quad & \int_{-\pi}^{\pi} \cos nx \cos mx \, dx = 0 & (n \neq m) \\ (b) \quad & \int_{-\pi}^{\pi} \sin nx \sin mx \, dx = 0 & (n \neq m) \\ (c) \quad & \int_{-\pi}^{\pi} \sin nx \cos mx \, dx = 0 & (n \neq m \text{ or } n = m). \end{aligned}$$

PROOF This follows simply by transforming the integrands trigonometrically from products into sums. In (9a) and (9b), by (11) in App. A3.1,

$$\begin{aligned} \int_{-\pi}^{\pi} \cos nx \cos mx \, dx &= \frac{1}{2} \int_{-\pi}^{\pi} \cos (n+m)x \, dx + \frac{1}{2} \int_{-\pi}^{\pi} \cos (n-m)x \, dx \\ \int_{-\pi}^{\pi} \sin nx \sin mx \, dx &= \frac{1}{2} \int_{-\pi}^{\pi} \cos (n-m)x \, dx - \frac{1}{2} \int_{-\pi}^{\pi} \cos (n+m)x \, dx. \end{aligned}$$

Since $m \neq n$ (integer!), the integrals on the right are all 0. Similarly, in (9c), for all integer m and n (without exception; do you see why?)

$$\int_{-\pi}^{\pi} \sin nx \cos mx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} \sin (n+m)x \, dx + \frac{1}{2} \int_{-\pi}^{\pi} \sin (n-m)x \, dx = 0 + 0. \quad \blacksquare$$

Application of Theorem 1 to the Fourier Series (5)

We prove (6.0). Integrating on both sides of (5) from $-\pi$ to π , we get

$$\int_{-\pi}^{\pi} f(x) \, dx = \int_{-\pi}^{\pi} \left[a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \right] dx.$$

We now assume that termwise integration is allowed. (We shall say in the proof of Theorem 2 when this is true.) Then we obtain

$$\int_{-\pi}^{\pi} f(x) \, dx = a_0 \int_{-\pi}^{\pi} dx + \sum_{n=1}^{\infty} \left(a_n \int_{-\pi}^{\pi} \cos nx \, dx + b_n \int_{-\pi}^{\pi} \sin nx \, dx \right).$$

The first term on the right equals $2\pi a_0$. Integration shows that all the other integrals are 0. Hence division by 2π gives (6.0).

We prove (6a). Multiplying (5) on both sides by $\cos mx$ with any **fixed** positive integer m and integrating from $-\pi$ to π , we have

$$(10) \quad \int_{-\pi}^{\pi} f(x) \cos mx \, dx = \int_{-\pi}^{\pi} \left[a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \right] \cos mx \, dx.$$

We now integrate term by term. Then on the right we obtain an integral of $a_0 \cos mx$, which is 0; an integral of $a_n \cos nx \cos mx$, which is $a_n \pi$ for $n = m$ and 0 for $n \neq m$ by (9a); and an integral of $b_n \sin nx \cos mx$, which is 0 for all n and m by (9c). Hence the right side of (10) equals $a_m \pi$. Division by π gives (6a) (with m instead of n).

We finally prove (6b). Multiplying (5) on both sides by $\sin mx$ with any **fixed** positive integer m and integrating from $-\pi$ to π , we get

$$(11) \quad \int_{-\pi}^{\pi} f(x) \sin mx \, dx = \int_{-\pi}^{\pi} \left[a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \right] \sin mx \, dx.$$

Integrating term by term, we obtain on the right an integral of $a_0 \sin mx$, which is 0; an integral of $a_n \cos nx \sin mx$, which is 0 by (9c); and an integral of $b_n \sin nx \sin mx$, which is $b_n \pi$ if $n = m$ and 0 if $n \neq m$, by (9b). This implies (6b) (with n denoted by m). This completes the proof of the Euler formulas (6) for the Fourier coefficients. ■

Convergence and Sum of a Fourier Series

The class of functions that can be represented by Fourier series is surprisingly large and general. Sufficient conditions valid in most applications are as follows.

THEOREM 2

Representation by a Fourier Series

Let $f(x)$ be periodic with period 2π and piecewise continuous (see Sec. 6.1) in the interval $-\pi \leq x \leq \pi$. Furthermore, let $f(x)$ have a left-hand derivative and a right-hand derivative at each point of that interval. Then the Fourier series (5) of $f(x)$ [with coefficients (6)] converges. Its sum is $f(x)$, except at points x_0 where $f(x)$ is discontinuous. There the sum of the series is the average of the left- and right-hand limits² of $f(x)$ at x_0 .

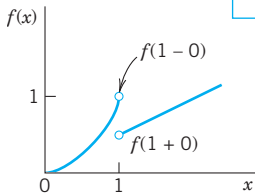


Fig. 262. Left- and right-hand limits

$$f(1-0) = 1,$$

$$f(1+0) = \frac{1}{2}$$

of the function

$$f(x) = \begin{cases} x^2 & \text{if } x < 1 \\ x/2 & \text{if } x \geq 1 \end{cases}$$

²The **left-hand limit** of $f(x)$ at x_0 is defined as the limit of $f(x)$ as x approaches x_0 from the left and is commonly denoted by $f(x_0 - 0)$. Thus

$$f(x_0 - 0) = \lim_{h \rightarrow 0} f(x_0 - h) \text{ as } h \rightarrow 0 \text{ through positive values.}$$

The **right-hand limit** is denoted by $f(x_0 + 0)$ and

$$f(x_0 + 0) = \lim_{h \rightarrow 0} f(x_0 + h) \text{ as } h \rightarrow 0 \text{ through positive values.}$$

The **left- and right-hand derivatives** of $f(x)$ at x_0 are defined as the limits of

$$\frac{f(x_0 - h) - f(x_0 - 0)}{-h} \quad \text{and} \quad \frac{f(x_0 + h) - f(x_0 + 0)}{-h},$$

respectively, as $h \rightarrow 0$ through positive values. Of course if $f(x)$ is continuous at x_0 , the last term in both numerators is simply $f(x_0)$.

PROOF We prove convergence, but only for a continuous function $f(x)$ having continuous first and second derivatives. And we do not prove that the sum of the series is $f(x)$ because these proofs are much more advanced; see, for instance, Ref. [C12] listed in App. 1. Integrating (6a) by parts, we obtain

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = \left. \frac{f(x) \sin nx}{n\pi} \right|_{-\pi}^{\pi} - \frac{1}{n\pi} \int_{-\pi}^{\pi} f'(x) \sin nx \, dx.$$

The first term on the right is zero. Another integration by parts gives

$$a_n = \left. \frac{f'(x) \cos nx}{n^2 \pi} \right|_{-\pi}^{\pi} - \frac{1}{n^2 \pi} \int_{-\pi}^{\pi} f''(x) \cos nx \, dx.$$

The first term on the right is zero because of the periodicity and continuity of $f'(x)$. Since f'' is continuous in the interval of integration, we have

$$|f''(x)| < M$$

for an appropriate constant M . Furthermore, $|\cos nx| \leq 1$. It follows that

$$|a_n| = \frac{1}{n^2 \pi} \left| \int_{-\pi}^{\pi} f''(x) \cos nx \, dx \right| < \frac{1}{n^2 \pi} \int_{-\pi}^{\pi} M \, dx = \frac{2M}{n^2}.$$

Similarly, $|b_n| < 2M/n^2$ for all n . Hence the absolute value of each term of the Fourier series of $f(x)$ is at most equal to the corresponding term of the series

$$|a_0| + 2M \left(1 + 1 + \frac{1}{2^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{3^2} + \cdots \right)$$

which is convergent. Hence that Fourier series converges and the proof is complete. (Readers already familiar with uniform convergence will see that, by the Weierstrass test in Sec. 15.5, under our present assumptions the Fourier series converges uniformly, and our derivation of (6) by integrating term by term is then justified by Theorem 3 of Sec. 15.5.) ■

EXAMPLE 2 Convergence at a Jump as Indicated in Theorem 2

The rectangular wave in Example 1 has a jump at $x = 0$. Its left-hand limit there is $-k$ and its right-hand limit is k (Fig. 261). Hence the average of these limits is 0. The Fourier series (8) of the wave does indeed converge to this value when $x = 0$ because then all its terms are 0. Similarly for the other jumps. This is in agreement with Theorem 2. ■

Summary. A Fourier series of a given function $f(x)$ of period 2π is a series of the form (5) with coefficients given by the Euler formulas (6). Theorem 2 gives conditions that are sufficient for this series to converge and at each x to have the value $f(x)$, except at discontinuities of $f(x)$, where the series equals the arithmetic mean of the left-hand and right-hand limits of $f(x)$ at that point.

PROBLEM SET 11.1

1–5 PERIOD, FUNDAMENTAL PERIOD

The *fundamental period* is the smallest positive period. Find it for

1. $\cos x$, $\sin x$, $\cos 2x$, $\sin 2x$, $\cos \pi x$, $\sin \pi x$,
 $\cos 2\pi x$, $\sin 2\pi x$
2. $\cos nx$, $\sin nx$, $\cos \frac{2\pi x}{k}$, $\sin \frac{2\pi x}{k}$, $\cos \frac{2\pi nx}{k}$,
 $\sin \frac{2\pi nx}{k}$
3. If $f(x)$ and $g(x)$ have period p , show that $h(x) = af(x) + bg(x)$ (a, b , constant) has the period p . Thus all functions of period p form a **vector space**.
4. **Change of scale.** If $f(x)$ has period p , show that $f(ax)$, $a \neq 0$, and $f(x/b)$, $b \neq 0$, are periodic functions of x of periods p/a and bp , respectively. Give examples.
5. Show that $f = \text{const}$ is periodic with any period but has no fundamental period.

6–10 GRAPHS OF 2π -PERIODIC FUNCTIONS

Sketch or graph $f(x)$ which for $-\pi < x < \pi$ is given as follows.

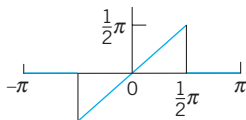
6. $f(x) = |x|$
7. $f(x) = |\sin x|$, $f(x) = \sin |x|$
8. $f(x) = e^{-|x|}$, $f(x) = |e^{-x}|$
9. $f(x) = \begin{cases} x & \text{if } -\pi < x < 0 \\ \pi - x & \text{if } 0 < x < \pi \end{cases}$
10. $f(x) = \begin{cases} -\cos^2 x & \text{if } -\pi < x < 0 \\ \cos^2 x & \text{if } 0 < x < \pi \end{cases}$

11. **Calculus review.** Review integration techniques for integrals as they are likely to arise from the Euler formulas, for instance, definite integrals of $x \cos nx$, $x^2 \sin nx$, $e^{-2x} \cos nx$, etc.

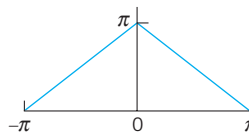
12–21 FOURIER SERIES

Find the Fourier series of the given function $f(x)$, which is assumed to have the period 2π . Show the details of your work. Sketch or graph the partial sums up to that including $\cos 5x$ and $\sin 5x$.

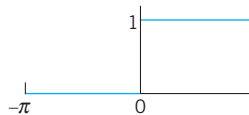
12. $f(x)$ in Prob. 6
13. $f(x)$ in Prob. 9
14. $f(x) = x^2$ ($-\pi < x < \pi$)
15. $f(x) = x^2$ ($0 < x < 2\pi$)
- 16.



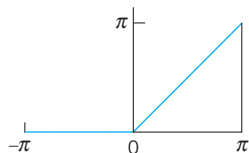
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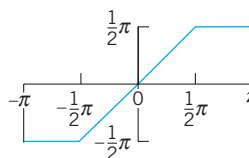
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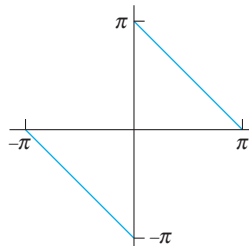
19.



20.



21.



22. **CAS EXPERIMENT. Graphing.** Write a program for graphing partial sums of the following series. Guess from the graph what $f(x)$ the series may represent. Confirm or disprove your guess by using the Euler formulas.

$$\begin{aligned} \text{(a)} \quad & 2(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \cdots) \\ & - 2(\frac{1}{2} \sin 2x + \frac{1}{4} \sin 4x + \frac{1}{6} \sin 6x \cdots) \\ \text{(b)} \quad & \frac{1}{2} + \frac{4}{\pi^2} \left(\cos x + \frac{1}{9} \cos 3x + \frac{1}{25} \cos 5x + \cdots \right) \\ \text{(c)} \quad & \frac{2}{3} \pi^2 + 4(\cos x - \frac{1}{4} \cos 2x + \frac{1}{9} \cos 3x - \frac{1}{16} \cos 4x \\ & + \cdots) \end{aligned}$$

23. **Discontinuities.** Verify the last statement in Theorem 2 for the discontinuities of $f(x)$ in Prob. 21.

24. **CAS EXPERIMENT. Orthogonality.** Integrate and graph the integral of the product $\cos mx \cos nx$ (with various integer m and n of your choice) from $-a$ to a as a function of a and conclude orthogonality of $\cos mx$

and $\cos nx$ ($m \neq n$) for $a = \pi$ from the graph. For what m and n will you get orthogonality for $a = \pi/2, \pi/3, \pi/4$? Other a ? Extend the experiment to $\cos mx \sin nx$ and $\sin mx \sin nx$.

25. CAS EXPERIMENT. Order of Fourier Coefficients.

The order seems to be $1/n$ if f is discontinuous, and $1/n^2$

if f is continuous but $f' = df/dx$ is discontinuous, $1/n^3$ if f and f' are continuous but f'' is discontinuous, etc. Try to verify this for examples. Try to prove it by integrating the Euler formulas by parts. What is the practical significance of this?

11.2 Arbitrary Period. Even and Odd Functions. Half-Range Expansions

We now expand our initial basic discussion of Fourier series.

Orientation. This section concerns three topics:

1. Transition from period 2π to any period $2L$, for the function f , simply by a transformation of scale on the x -axis.
2. Simplifications. Only cosine terms if f is even ("Fourier cosine series"). Only sine terms if f is odd ("Fourier sine series").
3. Expansion of f given for $0 \leq x \leq L$ in two Fourier series, one having only cosine terms and the other only sine terms ("half-range expansions").

1. From Period 2π to Any Period $p = 2L$

Clearly, periodic functions in applications may have any period, not just 2π as in the last section (chosen to have simple formulas). The notation $p = 2L$ for the period is practical because L will be a length of a violin string in Sec. 12.2, of a rod in heat conduction in Sec. 12.5, and so on.

The transition from period 2π to be period $p = 2L$ is effected by a suitable change of scale, as follows. Let $f(x)$ have period $p = 2L$. Then we can introduce a new variable v such that $f(x)$, as a function of v , has period 2π . If we set

$$(1) \quad (a) \quad x = \frac{p}{2\pi} v, \quad \text{so that} \quad (b) \quad v = \frac{2\pi}{p} x = \frac{\pi}{L} x$$

then $v = \pm\pi$ corresponds to $x = \pm L$. This means that f , as a function of v , has period 2π and, therefore, a Fourier series of the form

$$(2) \quad f(x) = f\left(\frac{L}{\pi} v\right) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nv + b_n \sin nv)$$

with coefficients obtained from (6) in the last section

$$(3) \quad \begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f\left(\frac{L}{\pi} v\right) dv, & a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f\left(\frac{L}{\pi} v\right) \cos nv \, dv, \\ b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f\left(\frac{L}{\pi} v\right) \sin nv \, dv. \end{aligned}$$

We could use these formulas directly, but the change to x simplifies calculations. Since

$$(4) \quad v = \frac{\pi}{L}x, \quad \text{we have} \quad dv = \frac{\pi}{L} dx$$

and we integrate over x from $-L$ to L . Consequently, we obtain for a function $f(x)$ of period $2L$ the Fourier series

$$(5) \quad f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{L}x + b_n \sin \frac{n\pi}{L}x \right)$$

with the **Fourier coefficients** of $f(x)$ given by the **Euler formulas** (π/L in dx cancels $1/\pi$ in (3))

$$(6) \quad \begin{aligned} (0) \quad a_0 &= \frac{1}{2L} \int_{-L}^L f(x) dx \\ (a) \quad a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx & n = 1, 2, \dots \\ (b) \quad b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx & n = 1, 2, \dots \end{aligned}$$

Just as in Sec. 11.1, we continue to call (5) with any coefficients a **trigonometric series**. And we can integrate from 0 to $2L$ or over any other interval of length $p = 2L$.

EXAMPLE 1 Periodic Rectangular Wave

Find the Fourier series of the function (Fig. 263)

$$f(x) = \begin{cases} 0 & \text{if } -2 < x < -1 \\ k & \text{if } -1 < x < 1 \\ 0 & \text{if } 1 < x < 2 \end{cases} \quad p = 2L = 4, \quad L = 2.$$

Solution. From (6.0) we obtain $a_0 = k/2$ (verify!). From (6a) we obtain

$$a_n = \frac{1}{2} \int_{-2}^2 f(x) \cos \frac{n\pi x}{2} dx = \frac{1}{2} \int_{-1}^1 k \cos \frac{n\pi x}{2} dx = \frac{2k}{n\pi} \sin \frac{n\pi}{2}.$$

Thus $a_n = 0$ if n is even and

$$a_n = 2k/n\pi \quad \text{if } n = 1, 5, 9, \dots, \quad a_n = -2k/n\pi \quad \text{if } n = 3, 7, 11, \dots.$$

From (6b) we find that $b_n = 0$ for $n = 1, 2, \dots$. Hence the Fourier series is a **Fourier cosine series** (that is, it has no sine terms)

$$f(x) = \frac{k}{2} + \frac{2k}{\pi} \left(\cos \frac{\pi}{2}x - \frac{1}{3} \cos \frac{3\pi}{2}x + \frac{1}{5} \cos \frac{5\pi}{2}x - \dots \right).$$

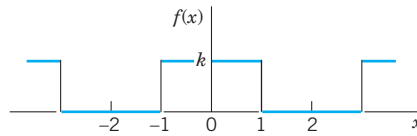


Fig. 263. Example 1

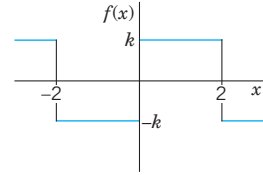


Fig. 264. Example 2

EXAMPLE 2 Periodic Rectangular Wave. Change of Scale

Find the Fourier series of the function (Fig. 264)

$$f(x) = \begin{cases} -k & \text{if } -2 < x < 0 \\ k & \text{if } 0 < x < 2 \end{cases} \quad p = 2L = 4, \quad L = 2.$$

Solution. Since $L = 2$, we have in (3) $v = \pi x/2$ and obtain from (8) in Sec. 11.1 with v instead of x , that is,

$$g(v) = \frac{4k}{\pi} \left(\sin v + \frac{1}{3} \sin 3v + \frac{1}{5} \sin 5v + \cdots \right)$$

the present Fourier series

$$f(x) = \frac{4k}{\pi} \left(\sin \frac{\pi}{2}x + \frac{1}{3} \sin \frac{3\pi}{2}x + \frac{1}{5} \sin \frac{5\pi}{2}x + \cdots \right).$$

Confirm this by using (6) and integrating. ■

EXAMPLE 3 Half-Wave Rectifier

A sinusoidal voltage $E \sin \omega t$, where t is time, is passed through a half-wave rectifier that clips the negative portion of the wave (Fig. 265). Find the Fourier series of the resulting periodic function

$$u(t) = \begin{cases} 0 & \text{if } -L < t < 0, \\ E \sin \omega t & \text{if } 0 < t < L \end{cases} \quad p = 2L = \frac{2\pi}{\omega}, \quad L = \frac{\pi}{\omega}.$$

Solution. Since $u = 0$ when $-L < t < 0$, we obtain from (6.0), with t instead of x ,

$$a_0 = \frac{\omega}{2\pi} \int_0^{\pi/\omega} E \sin \omega t \, dt = \frac{E}{\pi}$$

and from (6a), by using formula (11) in App. A3.1 with $x = \omega t$ and $y = n\omega t$,

$$a_n = \frac{\omega}{\pi} \int_0^{\pi/\omega} E \sin \omega t \cos n\omega t \, dt = \frac{\omega E}{2\pi} \int_0^{\pi/\omega} [\sin(1+n)\omega t + \sin(1-n)\omega t] \, dt.$$

If $n = 1$, the integral on the right is zero, and if $n = 2, 3, \dots$, we readily obtain

$$\begin{aligned} a_n &= \frac{\omega E}{2\pi} \left[-\frac{\cos(1+n)\omega t}{(1+n)\omega} - \frac{\cos(1-n)\omega t}{(1-n)\omega} \right]_0^{\pi/\omega} \\ &= \frac{E}{2\pi} \left(\frac{-\cos(1+n)\pi + 1}{1+n} + \frac{-\cos(1-n)\pi + 1}{1-n} \right). \end{aligned}$$

If n is odd, this is equal to zero, and for even n we have

$$a_n = \frac{E}{2\pi} \left(\frac{2}{1+n} + \frac{2}{1-n} \right) = -\frac{2E}{(n-1)(n+1)\pi} \quad (n = 2, 4, \dots).$$

In a similar fashion we find from (6b) that $b_1 = E/2$ and $b_n = 0$ for $n = 2, 3, \dots$. Consequently,

$$u(t) = \frac{E}{\pi} + \frac{E}{2} \sin \omega t - \frac{2E}{\pi} \left(\frac{1}{1 \cdot 3} \cos 2\omega t + \frac{1}{3 \cdot 5} \cos 4\omega t + \dots \right).$$

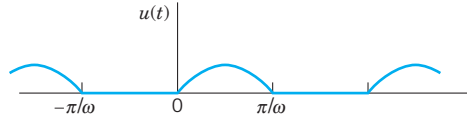


Fig. 265. Half-wave rectifier

2. Simplifications: Even and Odd Functions

If $f(x)$ is an **even function**, that is, $f(-x) = f(x)$ (see Fig. 266), its Fourier series (5) reduces to a **Fourier cosine series**

$$(5^*) \quad f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{L} x \quad (f \text{ even})$$

with coefficients (note: integration from 0 to L only!)

$$(6^*) \quad a_0 = \frac{1}{L} \int_0^L f(x) dx, \quad a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx, \quad n = 1, 2, \dots$$

If $f(x)$ is an **odd function**, that is, $f(-x) = -f(x)$ (see Fig. 267), its Fourier series (5) reduces to a **Fourier sine series**

$$(5^{**}) \quad f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{L} x \quad (f \text{ odd})$$

with coefficients

$$(6^{**}) \quad b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx.$$

These formulas follow from (5) and (6) by remembering from calculus that the definite integral gives the net area (= area above the axis minus area below the axis) under the curve of a function between the limits of integration. This implies

$$(7) \quad \begin{aligned} (a) \quad \int_{-L}^L g(x) dx &= 2 \int_0^L g(x) dx && \text{for even } g \\ (b) \quad \int_{-L}^L h(x) dx &= 0 && \text{for odd } h \end{aligned}$$

Formula (7b) implies the reduction to the cosine series (even f makes $f(x) \sin(n\pi x/L)$ odd since $\sin$ is odd) and to the sine series (odd f makes $f(x) \cos(n\pi x/L)$ odd since $\cos$ is even). Similarly, (7a) reduces the integrals in (6*) and (6**) to integrals from 0 to L . These reductions are obvious from the graphs of an even and an odd function. (Give a formal proof.)

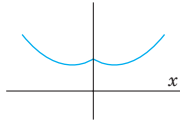


Fig. 266.

Even function

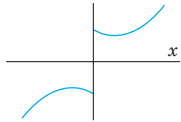


Fig. 267.

Odd function

Summary

Even Function of Period 2π . If f is even and $L = \pi$, then

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

with coefficients

$$a_0 = \frac{1}{\pi} \int_0^{\pi} f(x) dx, \quad a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx, \quad n = 1, 2, \dots$$

Odd Function of Period 2π . If f is odd and $L = \pi$, then

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

with coefficients

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx, \quad n = 1, 2, \dots$$

EXAMPLE 4 Fourier Cosine and Sine Series

The rectangular wave in Example 1 is even. Hence it follows without calculation that its Fourier series is a Fourier cosine series, the b_n are all zero. Similarly, it follows that the Fourier series of the odd function in Example 2 is a Fourier sine series.

In Example 3 you can see that the Fourier cosine series represents $u(t) = E/\pi - \frac{1}{2}E \sin \omega t$. Can you prove that this is an even function? ■

Further simplifications result from the following property, whose very simple proof is left to the student.

THEOREM 1**Sum and Scalar Multiple**

The Fourier coefficients of a sum $f_1 + f_2$ are the sums of the corresponding Fourier coefficients of f_1 and f_2 .

The Fourier coefficients of cf are c times the corresponding Fourier coefficients of f .

EXAMPLE 5 Sawtooth Wave

Find the Fourier series of the function (Fig. 268)

$$f(x) = x + \pi \quad \text{if} \quad -\pi < x < \pi \quad \text{and} \quad f(x + 2\pi) = f(x).$$

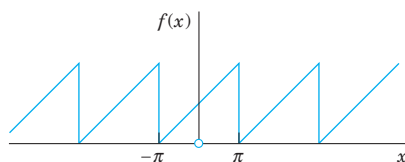


Fig. 268. The function $f(x)$. Sawtooth wave

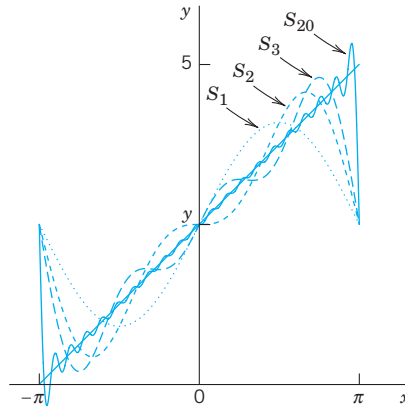


Fig. 269. Partial sums S_1, S_2, S_3, S_{20} in Example 5

Solution. We have $f = f_1 + f_2$, where $f_1 = x$ and $f_2 = \pi$. The Fourier coefficients of f_2 are zero, except for the first one (the constant term), which is π . Hence, by Theorem 1, the Fourier coefficients a_n, b_n are those of f_1 , except for a_0 , which is π . Since f_1 is odd, $a_n = 0$ for $n = 1, 2, \dots$, and

$$b_n = \frac{2}{\pi} \int_0^{\pi} f_1(x) \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} x \sin nx \, dx.$$

Integrating by parts, we obtain

$$b_n = \frac{2}{\pi} \left[\frac{-x \cos nx}{n} \Big|_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos nx \, dx \right] = -\frac{2}{n} \cos n\pi.$$

Hence $b_1 = 2, b_2 = -\frac{2}{2}, b_3 = \frac{2}{3}, b_4 = -\frac{2}{4}, \dots$, and the Fourier series of $f(x)$ is

$$f(x) = \pi + 2 \left(\sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x - \dots \right). \quad (\text{Fig. 269}) \quad \blacksquare$$

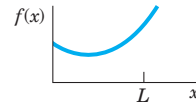
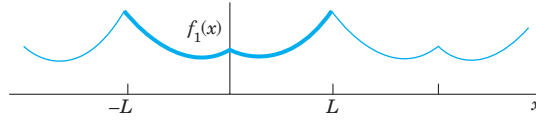
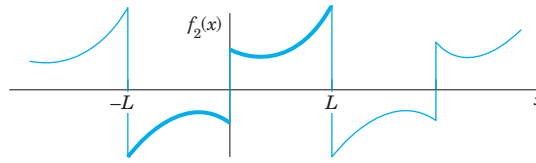
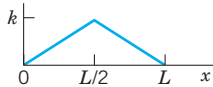
3. Half-Range Expansions

Half-range expansions are Fourier series. The idea is simple and useful. Figure 270 explains it. We want to represent $f(x)$ in Fig. 270.0 by a Fourier series, where $f(x)$ may be the shape of a distorted violin string or the temperature in a metal bar of length L , for example. (Corresponding problems will be discussed in Chap. 12.) Now comes the idea.

We could extend $f(x)$ as a function of period L and develop the extended function into a Fourier series. But this series would, in general, contain *both* cosine *and* sine terms. We can do better and get simpler series. Indeed, for our given f we can calculate Fourier coefficients from (6*) or from (6**). And we have a choice and can take what seems more practical. If we use (6*), we get (5*). This is the **even periodic extension** f_1 of f in Fig. 270a. If we choose (6**) instead, we get (5**), the **odd periodic extension** f_2 of f in Fig. 270b.

Both extensions have period $2L$. This motivates the name **half-range expansions**: f is given (and of physical interest) only on half the range, that is, on half the interval of periodicity of length $2L$.

Let us illustrate these ideas with an example that we shall also need in Chap. 12.

(0) The given function $f(x)$ (a) $f(x)$ continued as an **even** periodic function of period $2L$ (b) $f(x)$ continued as an **odd** periodic function of period $2L$ **Fig. 270.** Even and odd extensions of period $2L$ **EXAMPLE 6** “Triangle” and Its Half-Range Expansions**Fig. 271.** The given function in Example 6

Find the two half-range expansions of the function (Fig. 271)

$$f(x) = \begin{cases} \frac{2k}{L}x & \text{if } 0 < x < \frac{L}{2} \\ \frac{2k}{L}(L-x) & \text{if } \frac{L}{2} < x < L. \end{cases}$$

Solution. (a) *Even periodic extension.* From (6*) we obtain

$$a_0 = \frac{1}{L} \left[\frac{2k}{L} \int_0^{L/2} x \, dx + \frac{2k}{L} \int_{L/2}^L (L-x) \, dx \right] = \frac{k}{2},$$

$$a_n = \frac{2}{L} \left[\frac{2k}{L} \int_0^{L/2} x \cos \frac{n\pi}{L} x \, dx + \frac{2k}{L} \int_{L/2}^L (L-x) \cos \frac{n\pi}{L} x \, dx \right].$$

We consider a_n . For the first integral we obtain by integration by parts

$$\begin{aligned} \int_0^{L/2} x \cos \frac{n\pi}{L} x \, dx &= \frac{Lx}{n\pi} \sin \frac{n\pi}{L} x \Big|_0^{L/2} - \frac{L}{n\pi} \int_0^{L/2} \sin \frac{n\pi}{L} x \, dx \\ &= \frac{L^2}{2n\pi} \sin \frac{n\pi}{2} + \frac{L^2}{n^2\pi^2} \left(\cos \frac{n\pi}{2} - 1 \right). \end{aligned}$$

Similarly, for the second integral we obtain

$$\begin{aligned} \int_{L/2}^L (L-x) \cos \frac{n\pi}{L} x \, dx &= \frac{L}{n\pi} (L-x) \sin \frac{n\pi}{L} x \Big|_{L/2}^L + \frac{L}{n\pi} \int_{L/2}^L \sin \frac{n\pi}{L} x \, dx \\ &= \left(0 - \frac{L}{n\pi} \left(L - \frac{L}{2} \right) \sin \frac{n\pi}{2} \right) - \frac{L^2}{n^2\pi^2} \left(\cos n\pi - \cos \frac{n\pi}{2} \right). \end{aligned}$$

We insert these two results into the formula for a_n . The sine terms cancel and so does a factor L^2 . This gives

$$a_n = \frac{4k}{n^2\pi^2} \left(2 \cos \frac{n\pi}{2} - \cos n\pi - 1 \right).$$

Thus,

$$a_2 = -16k/(2^2\pi^2), \quad a_6 = -16k/(6^2\pi^2), \quad a_{10} = -16k/(10^2\pi^2), \dots$$

and $a_n = 0$ if $n \neq 2, 6, 10, 14, \dots$. Hence the first half-range expansion of $f(x)$ is (Fig. 272a)

$$f(x) = \frac{k}{2} - \frac{16k}{\pi^2} \left(\frac{1}{2^2} \cos \frac{2\pi}{L}x + \frac{1}{6^2} \cos \frac{6\pi}{L}x + \dots \right).$$

This Fourier cosine series represents the even periodic extension of the given function $f(x)$, of period $2L$.

(b) **Odd periodic extension.** Similarly, from (6**) we obtain

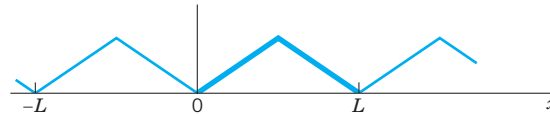
$$(5) \quad b_n = \frac{8k}{n^2\pi^2} \sin \frac{n\pi}{2}.$$

Hence the other half-range expansion of $f(x)$ is (Fig. 272b)

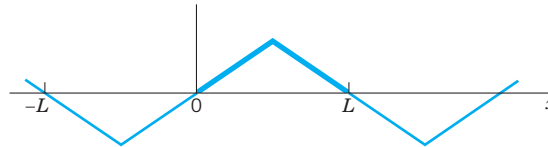
$$f(x) = \frac{8k}{\pi^2} \left(\frac{1}{1^2} \sin \frac{\pi}{L}x - \frac{1}{3^2} \sin \frac{3\pi}{L}x + \frac{1}{5^2} \sin \frac{5\pi}{L}x - \dots \right).$$

The series represents the odd periodic extension of $f(x)$, of period $2L$.

Basic applications of these results will be shown in Secs. 12.3 and 12.5. ■



(a) Even extension



(b) Odd extension

Fig. 272. Periodic extensions of $f(x)$ in Example 6

PROBLEM SET 11.2

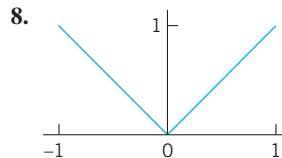
1–7 EVEN AND ODD FUNCTIONS

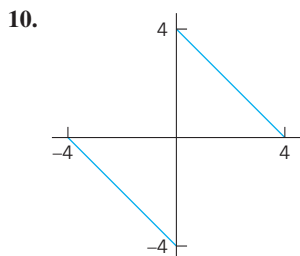
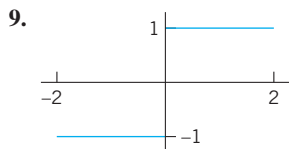
Are the following functions even or odd or neither even nor odd?

- e^x , $e^{-|x|}$, $x^3 \cos nx$, $x^2 \tan \pi x$, $\sinh x - \cosh x$
- $\sin^2 x$, $\sin(x^2)$, $\ln x$, $x/(x^2 + 1)$, $x \cot x$
- Sums and products of even functions
- Sums and products of odd functions
- Absolute values of odd functions
- Product of an odd times an even function
- Find all functions that are both even and odd.

8–17 FOURIER SERIES FOR PERIOD $p = 2L$

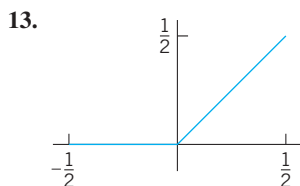
Is the given function even or odd or neither even nor odd? Find its Fourier series. Show details of your work.



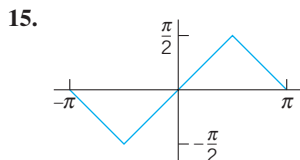


11. $f(x) = x^2$ $(-1 < x < 1)$, $p = 2$

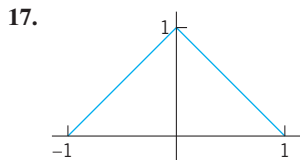
12. $f(x) = 1 - x^2/4$ $(-2 < x < 2)$, $p = 4$



14. $f(x) = \cos \pi x$ $(-\frac{1}{2} < x < \frac{1}{2})$, $p = 1$



16. $f(x) = x|x|$ $(-1 < x < 1)$, $p = 2$



18. **Rectifier.** Find the Fourier series of the function obtained by passing the voltage $v(t) = V_0 \cos 100\pi t$ through a half-wave rectifier that clips the negative half-waves.

19. **Trigonometric Identities.** Show that the familiar identities $\cos^3 x = \frac{3}{4} \cos x + \frac{1}{4} \cos 3x$ and $\sin^3 x = \frac{3}{4} \sin x - \frac{1}{4} \sin 3x$ can be interpreted as Fourier series expansions. Develop $\cos^4 x$.

20. **Numeric Values.** Using Prob. 11, show that $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \frac{1}{6} \pi^2$.

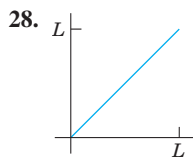
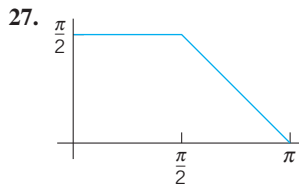
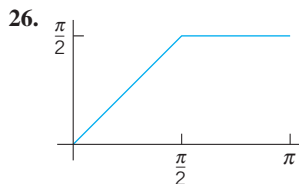
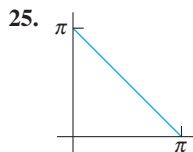
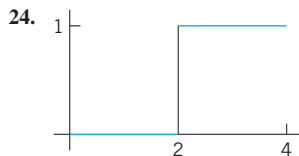
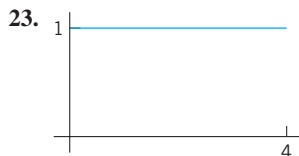
21. **CAS PROJECT. Fourier Series of 2L-Periodic Functions.** (a) Write a program for obtaining partial sums of a Fourier series (5).

(b) Apply the program to Probs. 8–11, graphing the first few partial sums of each of the four series on common axes. Choose the first five or more partial sums until they approximate the given function reasonably well. Compare and comment.

22. Obtain the Fourier series in Prob. 8 from that in Prob. 17.

23–29 HALF-RANGE EXPANSIONS

Find (a) the Fourier cosine series, (b) the Fourier sine series. Sketch $f(x)$ and its two periodic extensions. Show the details.



29. $f(x) = \sin x$ $(0 < x < \pi)$

30. Obtain the solution to Prob. 26 from that of Prob. 27.

11.3 Forced Oscillations

Fourier series have important applications for both ODEs and PDEs. In this section we shall focus on ODEs and cover similar applications for PDEs in Chap. 12. All these applications will show our indebtedness to Euler's and Fourier's ingenious idea of splitting up periodic functions into the simplest ones possible.

From Sec. 2.8 we know that forced oscillations of a body of mass m on a spring of modulus k are governed by the ODE

$$(1) \quad my'' + cy' + ky = r(t)$$

where $y = y(t)$ is the displacement from rest, c the damping constant, k the spring constant (spring modulus), and $r(t)$ the external force depending on time t . Figure 274 shows the model and Fig. 275 its electrical analog, an RLC -circuit governed by

$$(1^*) \quad LI'' + RI' + \frac{1}{C}I = E'(t) \quad (\text{Sec. 2.9}).$$

We consider (1). If $r(t)$ is a sine or cosine function and if there is damping ($c > 0$), then the steady-state solution is a harmonic oscillation with frequency equal to that of $r(t)$. However, if $r(t)$ is not a pure sine or cosine function but is any other periodic function, then the steady-state solution will be a superposition of harmonic oscillations with frequencies equal to that of $r(t)$ and integer multiples of these frequencies. And if one of these frequencies is close to the (practical) resonant frequency of the vibrating system (see Sec. 2.8), then the corresponding oscillation may be the dominant part of the response of the system to the external force. This is what the use of Fourier series will show us. Of course, this is quite surprising to an observer unfamiliar with Fourier series, which are highly important in the study of vibrating systems and resonance. Let us discuss the entire situation in terms of a typical example.

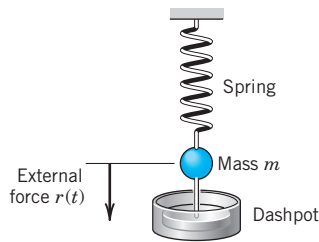


Fig. 274. Vibrating system under consideration

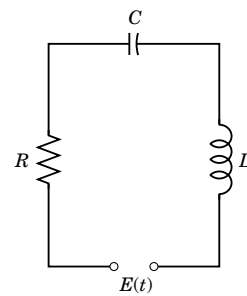


Fig. 275. Electrical analog of the system in Fig. 274 (RLC -circuit)

EXAMPLE 1

Forced Oscillations under a Nonsinusoidal Periodic Driving Force

In (1), let $m = 1$ (g), $c = 0.05$ (g/sec), and $k = 25$ (g/sec²), so that (1) becomes

$$(2) \quad y'' + 0.05y' + 25y = r(t)$$

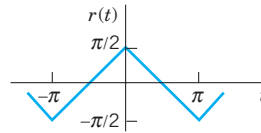


Fig. 276. Force in Example 1

where $r(t)$ is measured in $\text{g} \cdot \text{cm}/\text{sec}^2$. Let (Fig. 276)

$$r(t) = \begin{cases} t + \frac{\pi}{2} & \text{if } -\pi < t < 0, \\ -t + \frac{\pi}{2} & \text{if } 0 < t < \pi, \end{cases} \quad r(t + 2\pi) = r(t).$$

Find the steady-state solution $y(t)$.

Solution. We represent $r(t)$ by a Fourier series, finding

$$(3) \quad r(t) = \frac{4}{\pi} \left(\cos t + \frac{1}{3^2} \cos 3t + \frac{1}{5^2} \cos 5t + \cdots \right).$$

Then we consider the ODE

$$(4) \quad y'' + 0.05y' + 25y = \frac{4}{n^2\pi} \cos nt \quad (n = 1, 3, \dots)$$

whose right side is a single term of the series (3). From Sec. 2.8 we know that the steady-state solution $y_n(t)$ of (4) is of the form

$$(5) \quad y_n = A_n \cos nt + B_n \sin nt.$$

By substituting this into (4) we find that

$$(6) \quad A_n = \frac{4(25 - n^2)}{n^2\pi D_n}, \quad B_n = \frac{0.2}{n\pi D_n}, \quad \text{where} \quad D_n = (25 - n^2)^2 + (0.05n)^2.$$

Since the ODE (2) is linear, we may expect the steady-state solution to be

$$(7) \quad y = y_1 + y_3 + y_5 + \cdots$$

where y_n is given by (5) and (6). In fact, this follows readily by substituting (7) into (2) and using the Fourier series of $r(t)$, provided that termwise differentiation of (7) is permissible. (Readers already familiar with the notion of uniform convergence [Sec. 15.5] may prove that (7) may be differentiated term by term.)

From (6) we find that the amplitude of (5) is (a factor $\sqrt{D_n}$ cancels out)

$$C_n = \sqrt{A_n^2 + B_n^2} = \frac{4}{n^2\pi\sqrt{D_n}}.$$

Values of the first few amplitudes are

$$C_1 = 0.0531 \quad C_3 = 0.0088 \quad C_5 = 0.2037 \quad C_7 = 0.0011 \quad C_9 = 0.0003.$$

Figure 277 shows the input (multiplied by 0.1) and the output. For $n = 5$ the quantity D_n is very small, the denominator of C_5 is small, and C_5 is so large that y_5 is the dominating term in (7). Hence the output is almost a harmonic oscillation of five times the frequency of the driving force, a little distorted due to the term y_1 , whose amplitude is about 25% of that of y_5 . You could make the situation still more extreme by decreasing the damping constant c . Try it. ■

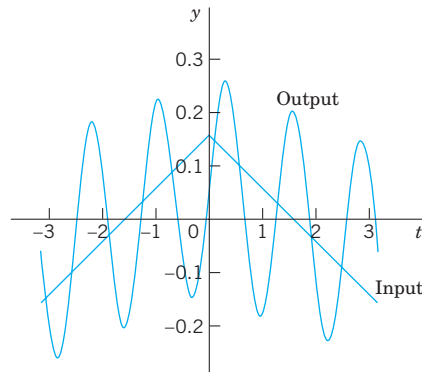


Fig. 277. Input and steady-state output in Example 1

PROBLEM SET 11.3

- Coefficients C_n .** Derive the formula for C_n from A_n and B_n .
- Change of spring and damping.** In Example 1, what happens to the amplitudes C_n if we take a stiffer spring, say, of $k = 49$? If we increase the damping?
- Phase shift.** Explain the role of the B_n 's. What happens if we let $c \rightarrow 0$?
- Differentiation of input.** In Example 1, what happens if we replace $r(t)$ with its derivative, the rectangular wave? What is the ratio of the new C_n to the old ones?
- Sign of coefficients.** Some of the A_n in Example 1 are positive, some negative. All B_n are positive. Is this physically understandable?

6–11 GENERAL SOLUTION

Find a general solution of the ODE $y'' + \omega^2 y = r(t)$ with $r(t)$ as given. Show the details of your work.

- $r(t) = \sin \alpha t + \sin \beta t$, $\omega^2 \neq \alpha^2, \beta^2$
- $r(t) = \sin t$, $\omega = 0.5, 0.9, 1.1, 1.5, 10$
- Rectifier.** $r(t) = \pi/4 |\cos t|$ if $-\pi < t < \pi$ and $r(t + 2\pi) = r(t)$, $|\omega| \neq 0, 2, 4, \dots$
- What kind of solution is excluded in Prob. 8 by $|\omega| \neq 0, 2, 4, \dots$?
- Rectifier.** $r(t) = \pi/4 |\sin t|$ if $0 < t < 2\pi$ and $r(t + 2\pi) = r(t)$, $|\omega| \neq 0, 2, 4, \dots$
- $r(t) = \begin{cases} -1 & \text{if } -\pi < t < 0 \\ 1 & \text{if } 0 < t < \pi \end{cases}$, $|\omega| \neq 1, 3, 5, \dots$
- CAS Program.** Write a program for solving the ODE just considered and for jointly graphing input and output of an initial value problem involving that ODE. Apply

the program to Probs. 7 and 11 with initial values of your choice.

13–16 STEADY-STATE DAMPED OSCILLATIONS

Find the steady-state oscillations of $y'' + cy' + y = r(t)$ with $c > 0$ and $r(t)$ as given. Note that the spring constant is $k = 1$. Show the details. In Probs. 14–16 sketch $r(t)$.

- $r(t) = \sum_{n=1}^N (a_n \cos nt + b_n \sin nt)$
- $r(t) = \begin{cases} -1 & \text{if } -\pi < t < 0 \\ 1 & \text{if } 0 < t < \pi \end{cases}$ and $r(t + 2\pi) = r(t)$
- $r(t) = t(\pi^2 - t^2)$ if $-\pi < t < \pi$ and $r(t + 2\pi) = r(t)$
- $r(t) = \begin{cases} t & \text{if } -\pi/2 < t < \pi/2 \\ \pi - t & \text{if } \pi/2 < t < 3\pi/2 \end{cases}$ and $r(t + 2\pi) = r(t)$

17–19 RLC-CIRCUIT

Find the steady-state current $I(t)$ in the RLC -circuit in Fig. 275, where $R = 10 \Omega$, $L = 1 \text{ H}$, $C = 10^{-1} \text{ F}$ and with $E(t)$ V as follows and periodic with period 2π . Graph or sketch the first four partial sums. Note that the coefficients of the solution decrease rapidly. *Hint.* Remember that the ODE contains $E'(t)$, not $E(t)$, cf. Sec. 2.9.

- $E(t) = \begin{cases} -50t^2 & \text{if } -\pi < t < 0 \\ 50t^2 & \text{if } 0 < t < \pi \end{cases}$

18. $E(t) = \begin{cases} 100(t - t^2) & \text{if } -\pi < t < 0 \\ 100(t + t^2) & \text{if } 0 < t < \pi \end{cases}$
19. $E(t) = 200t(\pi^2 - t^2) \quad (-\pi < t < \pi)$

20. CAS EXPERIMENT. Maximum Output Term. Graph and discuss outputs of $y'' + cy' + ky = r(t)$ with $r(t)$ as in Example 1 for various c and k with emphasis on the maximum C_n and its ratio to the second largest $|C_n|$.

11.4 Approximation by Trigonometric Polynomials

Fourier series play a prominent role not only in differential equations but also in **approximation theory**, an area that is concerned with approximating functions by other functions—usually simpler functions. Here is how Fourier series come into the picture.

Let $f(x)$ be a function on the interval $-\pi \leq x \leq \pi$ that can be represented on this interval by a Fourier series. Then the **N th partial sum** of the Fourier series

$$(1) \quad f(x) \approx a_0 + \sum_{n=1}^N (a_n \cos nx + b_n \sin nx)$$

is an approximation of the given $f(x)$. In (1) we choose an arbitrary N and keep it fixed. Then we ask whether (1) is the “best” approximation of f by a **trigonometric polynomial of the same degree N** , that is, by a function of the form

$$(2) \quad F(x) = A_0 + \sum_{n=1}^N (A_n \cos nx + B_n \sin nx) \quad (N \text{ fixed}).$$

Here, “best” means that the “error” of the approximation is as small as possible.

Of course we must first define what we mean by the **error** of such an approximation. We could choose the maximum of $|f(x) - F(x)|$. But in connection with Fourier series it is better to choose a definition of error that measures the goodness of agreement between f and F on the whole interval $-\pi \leq x \leq \pi$. This is preferable since the sum f of a Fourier series may have jumps: F in Fig. 278 is a good overall approximation of f , but the maximum of $|f(x) - F(x)|$ (more precisely, the *supremum*) is large. We choose

$$(3) \quad E = \int_{-\pi}^{\pi} (f - F)^2 dx.$$

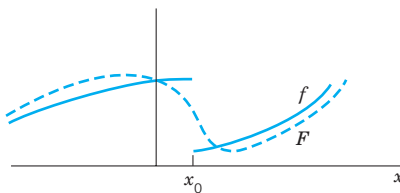


Fig. 278. Error of approximation

This is called the **square error** of F relative to the function f on the interval $-\pi \leq x \leq \pi$. Clearly, $E \geq 0$.

N being fixed, we want to determine the coefficients in (2) such that E is minimum. Since $(f - F)^2 = f^2 - 2fF + F^2$, we have

$$(4) \quad E = \int_{-\pi}^{\pi} f^2 dx - 2 \int_{-\pi}^{\pi} fF dx + \int_{-\pi}^{\pi} F^2 dx.$$

We square (2), insert it into the last integral in (4), and evaluate the occurring integrals. This gives integrals of $\cos^2 nx$ and $\sin^2 nx$ ($n \geq 1$), which equal π , and integrals of $\cos nx$, $\sin nx$, and $(\cos nx)(\sin mx)$, which are zero (just as in Sec. 11.1). Thus

$$\begin{aligned} \int_{-\pi}^{\pi} F^2 dx &= \int_{-\pi}^{\pi} \left[A_0 + \sum_{n=1}^N (A_n \cos nx + B_n \sin nx) \right]^2 dx \\ &= \pi(2A_0^2 + A_1^2 + \cdots + A_N^2 + B_1^2 + \cdots + B_N^2). \end{aligned}$$

We now insert (2) into the integral of fF in (4). This gives integrals of $f \cos nx$ as well as $f \sin nx$, just as in Euler's formulas, Sec. 11.1, for a_n and b_n (each multiplied by A_n or B_n). Hence

$$\int_{-\pi}^{\pi} fF dx = \pi(2A_0a_0 + A_1a_1 + \cdots + A_Na_N + B_1b_1 + \cdots + B_Nb_N).$$

With these expressions, (4) becomes

$$\begin{aligned} (5) \quad E &= \int_{-\pi}^{\pi} f^2 dx - 2\pi \left[2A_0a_0 + \sum_{n=1}^N (A_na_n + B_nb_n) \right] \\ &\quad + \pi \left[2A_0^2 + \sum_{n=1}^N (A_n^2 + B_n^2) \right]. \end{aligned}$$

We now take $A_n = a_n$ and $B_n = b_n$ in (2). Then in (5) the second line cancels half of the integral-free expression in the first line. Hence for this choice of the coefficients of F the square error, call it E^* , is

$$(6) \quad E^* = \int_{-\pi}^{\pi} f^2 dx - \pi \left[2a_0^2 + \sum_{n=1}^N (a_n^2 + b_n^2) \right].$$

We finally subtract (6) from (5). Then the integrals drop out and we get terms $A_n^2 - 2A_na_n + a_n^2 = (A_n - a_n)^2$ and similar terms $(B_n - b_n)^2$:

$$E - E^* = \pi \left\{ 2(A_0 - a_0)^2 + \sum_{n=1}^N [(A_n - a_n)^2 + (B_n - b_n)^2] \right\}.$$

Since the sum of squares of real numbers on the right cannot be negative,

$$E - E^* \geq 0, \quad \text{thus} \quad E \geq E^*,$$

and $E = E^*$ if and only if $A_0 = a_0, \dots, B_N = b_N$. This proves the following fundamental minimum property of the partial sums of Fourier series.

THEOREM 1

Minimum Square Error

The square error of F in (2) (with fixed N) relative to f on the interval $-\pi \leq x \leq \pi$ is minimum if and only if the coefficients of F in (2) are the Fourier coefficients of f . This minimum value E^* is given by (6).

From (6) we see that E^* cannot increase as N increases, but may decrease. Hence with increasing N the partial sums of the Fourier series of f yield better and better approximations to f , considered from the viewpoint of the square error.

Since $E^* \geq 0$ and (6) holds for every N , we obtain from (6) the important **Bessel's inequality**

$$(7) \quad 2a_0^2 + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)^2 dx$$

for the Fourier coefficients of any function f for which integral on the right exists. (For F. W. Bessel see Sec. 5.5.)

It can be shown (see [C12] in App. 1) that for such a function f , **Parseval's theorem** holds; that is, formula (7) holds with the equality sign, so that it becomes **Parseval's identity**³

$$(8) \quad 2a_0^2 + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)^2 dx.$$

EXAMPLE 1

Minimum Square Error for the Sawtooth Wave

Compute the minimum square error E^* of $F(x)$ with $N = 1, 2, \dots, 10, 20, \dots, 100$ and 1000 relative to

$$f(x) = x + \pi \quad (-\pi < x < \pi)$$

on the interval $-\pi \leq x \leq \pi$.

Solution. $F(x) = \pi + 2 \left(\sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x - \dots + \frac{(-1)^{N+1}}{N} \sin Nx \right)$ by Example 3 in Sec. 11.3. From this and (6),

$$E^* = \int_{-\pi}^{\pi} (x + \pi)^2 dx - \pi \left(2\pi^2 + 4 \sum_{n=1}^N \frac{1}{n^2} \right).$$

Numeric values are:

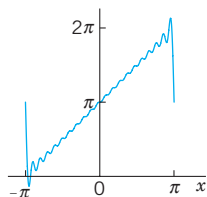


Fig. 279. F with $N = 20$ in Example 1

N	E^*	N	E^*	N	E^*	N	E^*
1	8.1045	6	1.9295	20	0.6129	70	0.1782
2	4.9629	7	1.6730	30	0.4120	80	0.1561
3	3.5666	8	1.4767	40	0.3103	90	0.1389
4	2.7812	9	1.3216	50	0.2488	100	0.1250
5	2.2786	10	1.1959	60	0.2077	1000	0.0126

³MARC ANTOINE PARSEVAL (1755–1836), French mathematician. A physical interpretation of the identity follows in the next section.

$F = S_1, S_2, S_3$ are shown in Fig. 269 in Sec. 11.2, and $F = S_{20}$ is shown in Fig. 279. Although $|f(x) - F(x)|$ is large at $\pm\pi$ (how large?), where f is discontinuous, F approximates f quite well on the whole interval, except near $\pm\pi$, where “waves” remain owing to the “Gibbs phenomenon,” which we shall discuss in the next section.

Can you think of functions f for which E^* decreases more quickly with increasing N ? ■

PROBLEM SET 11.4

- 1. CAS Problem.** Do the numeric and graphic work in Example 1 in the text.

2–5 MINIMUM SQUARE ERROR

Find the trigonometric polynomial $F(x)$ of the form (2) for which the square error with respect to the given $f(x)$ on the interval $-\pi < x < \pi$ is minimum. Compute the minimum value for $N = 1, 2, \dots, 5$ (or also for larger values if you have a CAS).

2. $f(x) = x \quad (-\pi < x < \pi)$

3. $f(x) = |x| \quad (-\pi < x < \pi)$

4. $f(x) = x^2 \quad (-\pi < x < \pi)$

5. $f(x) = \begin{cases} -1 & \text{if } -\pi < x < 0 \\ 1 & \text{if } 0 < x < \pi \end{cases}$

6. Why are the square errors in Prob. 5 substantially larger than in Prob. 3?

7. $f(x) = x^3 \quad (-\pi < x < \pi)$

8. $f(x) = |\sin x| \quad (-\pi < x < \pi)$, full-wave rectifier

9. **Monotonicity.** Show that the minimum square error (6) is a monotone decreasing function of N . How can you use this in practice?

10. **CAS EXPERIMENT. Size and Decrease of E^* .** Compare the size of the minimum square error E^* for functions of your choice. Find experimentally the

factors on which the decrease of E^* with N depends. For each function considered find the smallest N such that $E^* < 0.1$.

11–15 PARSEVAL'S IDENTITY

Using (8), prove that the series has the indicated sum. Compute the first few partial sums to see that the convergence is rapid.

11. $1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8} = 1.233700550$

Use Example 1 in Sec. 11.1.

12. $1 + \frac{1}{2^4} + \frac{1}{3^4} + \dots = \frac{\pi^4}{90} = 1.082323234$

Use Prob. 14 in Sec. 11.1.

13. $1 + \frac{1}{3^4} + \frac{1}{5^4} + \frac{1}{7^4} + \dots = \frac{\pi^4}{96} = 1.014678032$

Use Prob. 17 in Sec. 11.1.

14. $\int_{-\pi}^{\pi} \cos^4 x \, dx = \frac{3\pi}{4}$

15. $\int_{-\pi}^{\pi} \cos^6 x \, dx = \frac{5\pi}{8}$

11.5 Sturm–Liouville Problems. Orthogonal Functions

The idea of the Fourier series was to represent general periodic functions in terms of cosines and sines. The latter formed a *trigonometric system*. This trigonometric system has the desirable property of orthogonality which allows us to compute the coefficient of the Fourier series by the Euler formulas.

The question then arises, can this approach be generalized? That is, can we replace the trigonometric system of Sec. 11.1 by other *orthogonal systems* (*sets of other orthogonal functions*)? The answer is “yes” and will lead to generalized Fourier series, including the Fourier–Legendre series and the Fourier–Bessel series in Sec. 11.6.

To prepare for this generalization, we first have to introduce the concept of a Sturm–Liouville Problem. (The motivation for this approach will become clear as you read on.) Consider a second-order ODE of the form

$$(1) \quad [p(x)y']' + [q(x) + \lambda r(x)]y = 0$$

on some interval $a \leq x \leq b$, satisfying conditions of the form

$$(2) \quad \begin{aligned} (a) \quad & k_1 y + k_2 y' = 0 \quad \text{at } x = a \\ (b) \quad & l_1 y + l_2 y' = 0 \quad \text{at } x = b. \end{aligned}$$

Here λ is a parameter, and k_1, k_2, l_1, l_2 are given real constants. Furthermore, at least one of each constant in each condition (2) must be different from zero. (We will see in Example 1 that, if $p(x) = r(x) = 1$ and $q(x) = 0$, then $\sin \sqrt{\lambda}x$ and $\cos \sqrt{\lambda}x$ satisfy (1) and constants can be found to satisfy (2).) Equation (1) is known as a **Sturm–Liouville equation**.⁴ Together with conditions 2(a), 2(b) it is known as the **Sturm–Liouville problem**. It is an example of a boundary value problem.

A **boundary value problem** consists of an ODE and given boundary conditions referring to the two boundary points (endpoints) $x = a$ and $x = b$ of a given interval $a \leq x \leq b$.

The goal is to solve these type of problems. To do so, we have to consider

Eigenvalues, Eigenfunctions

Clearly, $y \equiv 0$ is a solution—the “**trivial solution**”—of the problem (1), (2) for any λ because (1) is homogeneous and (2) has zeros on the right. This is of no interest. We want to find **eigenfunctions** $y(x)$, that is, solutions of (1) satisfying (2) without being identically zero. We call a number λ for which an eigenfunction exists an **eigenvalue** of the Sturm–Liouville problem (1), (2).

Many important ODEs in engineering can be written as Sturm–Liouville equations. The following example serves as a case in point.

EXAMPLE 1 Trigonometric Functions as Eigenfunctions. Vibrating String

Find the eigenvalues and eigenfunctions of the Sturm–Liouville problem

$$(3) \quad y'' + \lambda y = 0, \quad y(0) = 0, \quad y(\pi) = 0.$$

This problem arises, for instance, if an elastic string (a violin string, for example) is stretched a little and fixed at its ends $x = 0$ and $x = \pi$ and then allowed to vibrate. Then $y(x)$ is the “space function” of the deflection $u(x, t)$ of the string, assumed in the form $u(x, t) = y(x)w(t)$, where t is time. (This model will be discussed in great detail in Secs. 12.2–12.4.)

Solution. From (1) and (2) we see that $p = 1, q = 0, r = 1$ in (1), and $a = 0, b = \pi, k_1 = l_1 = 1, k_2 = l_2 = 0$ in (2). For negative $\lambda = -v^2$ a general solution of the ODE in (3) is $y(x) = c_1 e^{vx} + c_2 e^{-vx}$. From the boundary conditions we obtain $c_1 = c_2 = 0$, so that $y \equiv 0$, which is not an eigenfunction. For $\lambda = 0$ the situation is similar. For positive $\lambda = v^2$ a general solution is

$$y(x) = A \cos vx + B \sin vx.$$

⁴JACQUES CHARLES FRANÇOIS STURM (1803–1855) was born and studied in Switzerland and then moved to Paris, where he later became the successor of Poisson in the chair of mechanics at the Sorbonne (the University of Paris).

JOSEPH LIOUVILLE (1809–1882), French mathematician and professor in Paris, contributed to various fields in mathematics and is particularly known by his important work in complex analysis (Liouville’s theorem; Sec. 14.4), special functions, differential geometry, and number theory.

From the first boundary condition we obtain $y(0) = A = 0$. The second boundary condition then yields

$$y(\pi) = B \sin v\pi = 0, \quad \text{thus} \quad v = 0, \pm 1, \pm 2, \dots$$

For $v = 0$ we have $y \equiv 0$. For $\lambda = v^2 = 1, 4, 9, 16, \dots$, taking $B = 1$, we obtain

$$y(x) = \sin vx \quad (v = \sqrt{\lambda} = 1, 2, \dots).$$

Hence the eigenvalues of the problem are $\lambda = v^2$, where $v = 1, 2, \dots$, and corresponding eigenfunctions are $y(x) = \sin vx$, where $v = 1, 2, \dots$. ■

Note that the solution to this problem is precisely the trigonometric system of the Fourier series considered earlier. It can be shown that, under rather general conditions on the functions p, q, r in (1), the Sturm–Liouville problem (1), (2) has infinitely many eigenvalues. The corresponding rather complicated theory can be found in Ref. [All] listed in App. 1.

Furthermore, if p, q, r , and p' in (1) are real-valued and continuous on the interval $a \leq x \leq b$ and r is positive throughout that interval (or negative throughout that interval), then all the eigenvalues of the Sturm–Liouville problem (1), (2) are real. (Proof in App. 4.) This is what the engineer would expect since eigenvalues are often related to frequencies, energies, or other physical quantities that must be real.

The most remarkable and important property of eigenfunctions of Sturm–Liouville problems is their **orthogonality**, which will be crucial in series developments in terms of eigenfunctions, as we shall see in the next section. This suggests that we should next consider orthogonal functions.

Orthogonal Functions

Functions $y_1(x), y_2(x), \dots$ defined on some interval $a \leq x \leq b$ are called **orthogonal** on this interval with respect to the **weight function** $r(x) > 0$ if for all m and all n different from m ,

$$(4) \quad (y_m, y_n) = \int_a^b r(x) y_m(x) y_n(x) dx = 0 \quad (m \neq n).$$

(y_m, y_n) is a **standard notation** for this integral. **The norm** $\|y_m\|$ of y_m is defined by

$$(5) \quad \|y_m\| = \sqrt{(y_m, y_m)} = \sqrt{\int_a^b r(x) y_m^2(x) dx}.$$

Note that this is the square root of the integral in (4) with $n = m$.

The functions $y_1, y_2, \dots$ are called **orthonormal** on $a \leq x \leq b$ if they are orthogonal on this interval and all have norm 1. Then we can write (4), (5) jointly by using the **Kronecker symbol**⁵ δ_{mn} , namely,

$$(y_m, y_n) = \int_a^b r(x) y_m(x) y_n(x) dx = \delta_{mn} = \begin{cases} 0 & \text{if } m \neq n \\ 1 & \text{if } m = n. \end{cases}$$

⁵LEOPOLD KRONECKER (1823–1891). German mathematician at Berlin University, who made important contributions to algebra, group theory, and number theory.

If $r(x) = 1$, we more briefly call the functions *orthogonal* instead of orthogonal with respect to $r(x) = 1$; similarly for orthonormality. Then

$$(y_m, y_n) = \int_a^b y_m(x) y_n(x) dx = 0 \quad (m \neq n), \quad \|y_m\| = \sqrt{(y_m, y_m)} = \sqrt{\int_a^b y_m^2(x) dx}.$$

The next example serves as an illustration of the material on orthogonal functions just discussed.

EXAMPLE 2 Orthogonal Functions. Orthonormal Functions. Notation

The functions $y_m(x) = \sin mx$, $m = 1, 2, \dots$ form an orthogonal set on the interval $-\pi \leq x \leq \pi$, because for $m \neq n$ we obtain by integration [see (11) in App. A3.1]

$$(y_m, y_n) = \int_{-\pi}^{\pi} \sin mx \sin nx dx = \frac{1}{2} \int_{-\pi}^{\pi} \cos(m-n)x dx - \frac{1}{2} \int_{-\pi}^{\pi} \cos(m+n)x dx = 0, \quad (m \neq n).$$

The norm $\|y_m\| = \sqrt{(y_m, y_m)}$ equals $\sqrt{\pi}$ because

$$\|y_m\|^2 = (y_m, y_m) = \int_{-\pi}^{\pi} \sin^2 mx dx = \pi \quad (m = 1, 2, \dots)$$

Hence the corresponding orthonormal set, obtained by division by the norm, is

$$\frac{\sin x}{\sqrt{\pi}}, \quad \frac{\sin 2x}{\sqrt{\pi}}, \quad \frac{\sin 3x}{\sqrt{\pi}}, \quad \dots$$

Theorem 1 shows that for any Sturm–Liouville problem, the eigenfunctions associated with these problems are orthogonal. This means, in practice, if we can formulate a problem as a Sturm–Liouville problem, then by this theorem we are guaranteed orthogonality.

THEOREM 1

Orthogonality of Eigenfunctions of Sturm–Liouville Problems

Suppose that the functions p , q , r , and p' in the Sturm–Liouville equation (1) are real-valued and continuous and $r(x) > 0$ on the interval $a \leq x \leq b$. Let $y_m(x)$ and $y_n(x)$ be eigenfunctions of the Sturm–Liouville problem (1), (2) that correspond to different eigenvalues λ_m and λ_n , respectively. Then y_m , y_n are orthogonal on that interval with respect to the weight function r , that is,

$$(6) \quad (y_m, y_n) = \int_a^b r(x) y_m(x) y_n(x) dx = 0 \quad (m \neq n).$$

If $p(a) = 0$, then (2a) can be dropped from the problem. If $p(b) = 0$, then (2b) can be dropped. [It is then required that y and y' remain bounded at such a point, and the problem is called **singular**, as opposed to a **regular problem** in which (2) is used.]

If $p(a) = p(b)$, then (2) can be replaced by the “**periodic boundary conditions**”

$$(7) \quad y(a) = y(b), \quad y'(a) = y'(b).$$

The boundary value problem consisting of the Sturm–Liouville equation (1) and the periodic boundary conditions (7) is called a **periodic Sturm–Liouville problem**.

PROOF By assumption, y_m and y_n satisfy the Sturm–Liouville equations

$$(py'_m)' + (q + \lambda_m r)y_m = 0$$

$$(py'_n)' + (q + \lambda_n r)y_n = 0$$

respectively. We multiply the first equation by y_n , the second by $-y_m$, and add,

$$(\lambda_m - \lambda_n)ry_m y_n = y_m(py'_n)' - y_n(py'_m)' = [(py'_n)y_m - (py'_m)y_n]'$$

where the last equality can be readily verified by performing the indicated differentiation of the last expression in brackets. This expression is continuous on $a \leq x \leq b$ since p and p' are continuous by assumption and y_m, y_n are solutions of (1). Integrating over x from a to b , we thus obtain

$$(8) \quad (\lambda_m - \lambda_n) \int_a^b r y_m y_n dx = [p(y'_n y_m - y'_m y_n)]_a^b \quad (a < b).$$

The expression on the right equals the sum of the subsequent Lines 1 and 2,

$$(9) \quad \begin{aligned} & p(b)[y'_n(b)y_m(b) - y'_m(b)y_n(b)] && \text{(Line 1)} \\ & -p(a)[y'_n(a)y_m(a) - y'_m(a)y_n(a)] && \text{(Line 2)}. \end{aligned}$$

Hence if (9) is zero, (8) with $\lambda_m - \lambda_n \neq 0$ implies the orthogonality (6). Accordingly, we have to show that (9) is zero, using the boundary conditions (2) as needed.

Case 1. $p(a) = p(b) = 0$. Clearly, (9) is zero, and (2) is not needed.

Case 2. $p(a) \neq 0, p(b) = 0$. Line 1 of (9) is zero. Consider Line 2. From (2a) we have

$$k_1 y_n(a) + k_2 y'_n(a) = 0,$$

$$k_1 y_m(a) + k_2 y'_m(a) = 0.$$

Let $k_2 \neq 0$. We multiply the first equation by $y_m(a)$, the last by $-y_n(a)$ and add,

$$k_2 [y'_n(a)y_m(a) - y'_m(a)y_n(a)] = 0.$$

This is k_2 times Line 2 of (9), which thus is zero since $k_2 \neq 0$. If $k_2 = 0$, then $k_1 \neq 0$ by assumption, and the argument of proof is similar.

Case 3. $p(a) = 0, p(b) \neq 0$. Line 2 of (9) is zero. From (2b) it follows that Line 1 of (9) is zero; this is similar to Case 2.

Case 4. $p(a) \neq 0, p(b) \neq 0$. We use both (2a) and (2b) and proceed as in Cases 2 and 3.

Case 5. $p(a) = p(b)$. Then (9) becomes

$$p(b)[y'_n(b)y_m(b) - y'_m(b)y_n(b) - y'_n(a)y_m(a) + y'_m(a)y_n(a)].$$

The expression in brackets $[\cdots]$ is zero, either by (2) used as before, or more directly by (7). Hence in this case, (7) can be used instead of (2), as claimed. This completes the proof of Theorem 1. ■

EXAMPLE 3 Application of Theorem 1. Vibrating String

The ODE in Example 1 is a Sturm–Liouville equation with $p = 1$, $q = 0$, and $r = 1$. From Theorem 1 it follows that the eigenfunctions $y_m = \sin mx$ ($m = 1, 2, \dots$) are orthogonal on the interval $0 \leq x \leq \pi$. ■

Example 3 confirms, from this new perspective, that the trigonometric system underlying the Fourier series is orthogonal, as we knew from Sec. 11.1.

EXAMPLE 4 Application of Theorem 1. Orthogonality of the Legendre Polynomials

Legendre's equation $(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$ may be written

$$[(1 - x^2)y']' + \lambda y = 0 \quad \lambda = n(n + 1).$$

Hence, this is a Sturm–Liouville equation (1) with $p = 1 - x^2$, $q = 0$, and $r = 1$. Since $p(-1) = p(1) = 0$, we need no boundary conditions, but have a “*singular*” *Sturm–Liouville problem* on the interval $-1 \leq x \leq 1$. We know that for $n = 0, 1, \dots$, hence $\lambda = 0, 1 \cdot 2, 2 \cdot 3, \dots$, the Legendre polynomials $P_n(x)$ are solutions of the problem. Hence these are the eigenfunctions. From Theorem 1 it follows that they are orthogonal on that interval, that is,

$$(10) \quad \int_{-1}^1 P_m(x)P_n(x) dx = 0 \quad (m \neq n). \quad \blacksquare$$

What we have seen is that the trigonometric system, underlying the Fourier series, is a solution to a Sturm–Liouville problem, as shown in Example 1, and that this trigonometric system is orthogonal, which we knew from Sec. 11.1 and confirmed in Example 3.

PROBLEM SET 11.5

- 1. Proof of Theorem 1.** Carry out the details in Cases 3 and 4.

2–6 ORTHOGONALITY

- 2. Normalization of eigenfunctions** y_m of (1), (2) means that we multiply y_m by a nonzero constant c_m such that c_my_m has norm 1. Show that $z_m = cy_m$ with *any* $c \neq 0$ is an eigenfunction for the eigenvalue corresponding to y_m .
- 3. Change of x .** Show that if the functions $y_0(x), y_1(x), \dots$ form an orthogonal set on an interval $a \leq x \leq b$ (with $r(x) = 1$), then the functions $y_0(ct + k), y_1(ct + k), \dots, c > 0$, form an orthogonal set on the interval $(a - k)/c \leq t \leq (b - k)/c$.
- 4. Change of x .** Using Prob. 3, derive the orthogonality of $1, \cos \pi x, \sin \pi x, \cos 2\pi x, \dots$ on $-1 \leq x \leq 1$ ($r(x) = 1$) from that of $1, \cos x, \sin x, \cos 2x, \sin 2x, \dots$ on $-\pi \leq x \leq \pi$.
- 5. Legendre polynomials.** Show that the functions $P_n(\cos \theta), n = 0, 1, \dots$, from an orthogonal set on the interval $0 \leq \theta \leq \pi$ with respect to the weight function $\sin \theta$.
- 6. Transformation to Sturm–Liouville form.** Show that $y'' + fy' + (g + \lambda h)y = 0$ takes the form (1) if you

set $p = \exp(\int f dx), q = pg, r = hp$. Why would you do such a transformation?

7–15 STURM–LIOUVILLE PROBLEMS

Find the eigenvalues and eigenfunctions. Verify orthogonality. Start by writing the ODE in the form (1), using Prob. 6. Show details of your work.

- 7.** $y'' + \lambda y = 0, \quad y(0) = 0, \quad y(10) = 0$
- 8.** $y'' + \lambda y = 0, \quad y(0) = 0, \quad y(L) = 0$
- 9.** $y'' + \lambda y = 0, \quad y(0) = 0, \quad y'(L) = 0$
- 10.** $y'' + \lambda y = 0, \quad y(0) = y(1), \quad y'(0) = y'(1)$
- 11.** $(y'/x)' + (\lambda + 1)y/x^3 = 0, \quad y(1) = 0, \quad y(e^\pi) = 0.$ (Set $x = e^t$.)
- 12.** $y'' - 2y' + (\lambda + 1)y = 0, \quad y(0) = 0, \quad y(1) = 0$
- 13.** $y'' + 8y' + (\lambda + 16)y = 0, \quad y(0) = 0, \quad y(\pi) = 0$
- 14. TEAM PROJECT. Special Functions. Orthogonal polynomials** play a great role in applications. For this reason, Legendre polynomials and various other orthogonal polynomials have been studied extensively; see Refs. [GenRef1], [GenRef10] in App. 1. Consider some of the most important ones as follows.

(a) **Chebyshev polynomials**⁶ of the first and second kind are defined by

$$T_n(x) = \cos(n \arccos x)$$

$$U_n(x) = \frac{\sin[(n+1) \arccos x]}{\sqrt{1-x^2}}$$

respectively, where $n = 0, 1, \dots$. Show that

$$\begin{aligned} T_0 &= 1, & T_1(x) &= x, & T_2(x) &= 2x^2 - 1, \\ & & T_3(x) &= 4x^3 - 3x, \\ U_0 &= 1, & U_1(x) &= 2x, & U_2(x) &= 4x^2 - 1, \\ & & U_3(x) &= 8x^3 - 4x. \end{aligned}$$

Show that the Chebyshev polynomials $T_n(x)$ are orthogonal on the interval $-1 \leq x \leq 1$ with respect to the weight function $r(x) = 1/\sqrt{1-x^2}$. (Hint. To evaluate the integral, set $\arccos x = \theta$.) Verify

that $T_n(x)$, $n = 0, 1, 2, 3$, satisfy the **Chebyshev equation**

$$(1-x^2)y'' - xy' + n^2y = 0.$$

(b) **Orthogonality on an infinite interval: Laguerre polynomials**⁷ are defined by $L_0 = 1$, and

$$L_n(x) = \frac{e^x}{n!} \frac{d^n(x^n e^{-x})}{dx^n}, \quad n = 1, 2, \dots$$

Show that

$$\begin{aligned} L_1(x) &= 1 - x, & L_2(x) &= 1 - 2x + x^2/2, \\ L_3(x) &= 1 - 3x + 3x^2/2 - x^3/6. \end{aligned}$$

Prove that the Laguerre polynomials are orthogonal on the positive axis $0 \leq x < \infty$ with respect to the weight function $r(x) = e^{-x}$. Hint. Since the highest power in L_m is x^m , it suffices to show that $\int e^{-x} x^k L_n dx = 0$ for $k < n$. Do this by k integrations by parts.

11.6 Orthogonal Series. Generalized Fourier Series

Fourier series are made up of the trigonometric system (Sec. 11.1), which is orthogonal, and orthogonality was essential in obtaining the Euler formulas for the Fourier coefficients. Orthogonality will also give us coefficient formulas for the desired generalized Fourier series, including the Fourier–Legendre series and the Fourier–Bessel series. This generalization is as follows.

Let $y_0, y_1, y_2, \dots$ be orthogonal with respect to a weight function $r(x)$ on an interval $a \leq x \leq b$, and let $f(x)$ be a function that can be represented by a convergent series

$$(1) \quad f(x) = \sum_{m=0}^{\infty} a_m y_m(x) = a_0 y_0(x) + a_1 y_1(x) + \dots$$

This is called an **orthogonal series**, **orthogonal expansion**, or **generalized Fourier series**. If the y_m are the eigenfunctions of a Sturm–Liouville problem, we call (1) an **eigenfunction expansion**. In (1) we use again m for summation since n will be used as a fixed order of Bessel functions.

Given $f(x)$, we have to determine the coefficients in (1), called the **Fourier constants** of $f(x)$ with respect to $y_0, y_1, \dots$. Because of the orthogonality, this is simple. Similarly to Sec. 11.1, we multiply both sides of (1) by $r(x)y_n(x)$ (n *fixed*) and then integrate on

⁶PAFNUTI CHEBYSHEV (1821–1894), Russian mathematician, is known for his work in approximation theory and the theory of numbers. Another transliteration of the name is TCHEBICHEF.

⁷EDMOND LAGUERRE (1834–1886), French mathematician, who did research work in geometry and in the theory of infinite series.

both sides from a to b . We assume that term-by-term integration is permissible. (This is justified, for instance, in the case of “uniform convergence,” as is shown in Sec. 15.5.) Then we obtain

$$(f, y_n) = \int_a^b r f y_n dx = \int_a^b r \left(\sum_{m=0}^{\infty} a_m y_m \right) y_n dx = \sum_{m=0}^{\infty} a_m \int_a^b r y_m y_n dx = \sum_{m=0}^{\infty} a_m (y_m, y_n).$$

Because of the orthogonality all the integrals on the right are zero, except when $m = n$. Hence the whole infinite series reduces to the single term

$$a_n (y_n, y_n) = a_n \|y_n\|^2. \quad \text{Thus} \quad (f, y_n) = a_n \|y_n\|^2.$$

Assuming that all the functions y_n have nonzero norm, we can divide by $\|y_n\|^2$; writing again m for n , to be in agreement with (1), we get the desired formula for the Fourier constants

$$(2) \quad a_m = \frac{(f, y_m)}{\|y_m\|^2} = \frac{1}{\|y_m\|^2} \int_a^b r(x) f(x) y_m(x) dx \quad (n = 0, 1, \dots).$$

This formula generalizes the Euler formulas (6) in Sec. 11.1 as well as the principle of their derivation, namely, by orthogonality.

EXAMPLE 1 Fourier–Legendre Series

A **Fourier–Legendre series** is an eigenfunction expansion

$$f(x) = \sum_{m=0}^{\infty} a_m P_m(x) = a_0 P_0 + a_1 P_1(x) + a_2 P_2(x) + \dots = a_0 + a_1 x + a_2 \left(\frac{3}{2} x^2 - \frac{1}{2} \right) + \dots$$

in terms of Legendre polynomials (Sec. 5.3). The latter are the eigenfunctions of the Sturm–Liouville problem in Example 4 of Sec. 11.5 on the interval $-1 \leq x \leq 1$. We have $r(x) = 1$ for Legendre’s equation, and (2) gives

$$(3) \quad a_m = \frac{2m+1}{2} \int_{-1}^1 f(x) P_m(x) dx, \quad m = 0, 1, \dots$$

because the norm is

$$(4) \quad \|P_m\| = \sqrt{\int_{-1}^1 P_m(x)^2 dx} = \sqrt{\frac{2}{2m+1}} \quad (m = 0, 1, \dots)$$

as we state without proof. The proof of (4) is tricky; it uses Rodrigues’s formula in Problem Set 5.2 and a reduction of the resulting integral to a quotient of gamma functions.

For instance, let $f(x) = \sin \pi x$. Then we obtain the coefficients

$$a_m = \frac{2m+1}{2} \int_{-1}^1 (\sin \pi x) P_m(x) dx, \quad \text{thus} \quad a_1 = \frac{3}{2} \int_{-1}^1 x \sin \pi x dx = \frac{3}{\pi} = 0.95493, \quad \text{etc.}$$

Hence the Fourier–Legendre series of $\sin \pi x$ is

$$\sin \pi x = 0.95493P_1(x) - 1.15824P_3(x) + 0.21929P_5(x) - 0.01664P_7(x) + 0.00068P_9(x) \\ - 0.00002P_{11}(x) + \cdots.$$

The coefficient of P_{13} is about $3 \cdot 10^{-7}$. The sum of the first three nonzero terms gives a curve that practically coincides with the sine curve. Can you see why the even-numbered coefficients are zero? Why a_3 is the absolutely biggest coefficient? ■

EXAMPLE 2 Fourier–Bessel Series

These series model vibrating membranes (Sec. 12.9) and other physical systems of circular symmetry. We derive these series in three steps.

Step 1. Bessel’s equation as a Sturm–Liouville equation. The Bessel function $J_n(x)$ with fixed integer $n \geq 0$ satisfies Bessel’s equation (Sec. 5.5)

$$\tilde{x}^2 \ddot{J}_n(\tilde{x}) + \tilde{x} \dot{J}_n(\tilde{x}) + (\tilde{x}^2 - n^2)J_n(\tilde{x}) = 0$$

where $\dot{J}_n = dJ_n/d\tilde{x}$ and $\ddot{J}_n = d^2J_n/d\tilde{x}^2$. We set $\tilde{x} = kx$. Then $x = \tilde{x}/k$ and by the chain rule, $\dot{J}_n = dJ_n/d\tilde{x} = (dJ_n/dx)/k$ and $\ddot{J}_n = J_n''/k^2$. In the first two terms of Bessel’s equation, k^2 and k drop out and we obtain

$$x^2 J_n''(kx) + x J_n'(kx) + (k^2 x^2 - n^2)J_n(kx) = 0.$$

Dividing by x and using $(xJ_n'(kx))' = xJ_n''(kx) + J_n'(kx)$ gives the Sturm–Liouville equation

$$(5) \quad [xJ_n'(kx)]' + \left(-\frac{n^2}{x} + \lambda x\right)J_n(kx) = 0 \quad \lambda = k^2$$

with $p(x) = x$, $q(x) = -n^2/x$, $r(x) = x$, and parameter $\lambda = k^2$. Since $p(0) = 0$, Theorem 1 in Sec. 11.5 implies orthogonality on an interval $0 \leq x \leq R$ (R given, fixed) of those solutions $J_n(kx)$ that are zero at $x = R$, that is,

$$(6) \quad J_n(kR) = 0 \quad (n \text{ fixed}).$$

Note that $q(x) = -n^2/x$ is discontinuous at 0, but this does not affect the proof of Theorem 1.

Step 2. Orthogonality. It can be shown (see Ref. [A13]) that $J_n(\tilde{x})$ has infinitely many zeros, say, $\tilde{x} = a_{n,1} < a_{n,2} < \cdots$ (see Fig. 110 in Sec. 5.4 for $n = 0$ and 1). Hence we must have

$$(7) \quad kR = \alpha_{n,m} \quad \text{thus} \quad k_{n,m} = \alpha_{n,m}/R \quad (m = 1, 2, \dots).$$

This proves the following orthogonality property.

THEOREM 1

Orthogonality of Bessel Functions

For each fixed nonnegative integer n the sequence of Bessel functions of the first kind $J_n(k_{n,1}x)$, $J_n(k_{n,2}x)$, $\cdots$ with $k_{n,m}$ as in (7) forms an orthogonal set on the interval $0 \leq x \leq R$ with respect to the weight function $r(x) = x$, that is,

$$(8) \quad \int_0^R x J_n(k_{n,m}x) J_n(k_{n,j}x) dx = 0 \quad (j \neq m, n \text{ fixed}).$$

Hence we have obtained *infinitely many orthogonal sets* of Bessel functions, one for each of $J_0, J_1, J_2, \dots$. Each set is orthogonal on an interval $0 \leq x \leq R$ with a fixed positive R of our choice and with respect to the weight x . The orthogonal set for J_n is $J_n(k_{n,1}x), J_n(k_{n,2}x), J_n(k_{n,3}x), \dots$, where n is fixed and $k_{n,m}$ is given by (7).

Step 3. Fourier–Bessel series. The Fourier–Bessel series corresponding to J_n (n fixed) is

$$(9) \quad f(x) = \sum_{m=1}^{\infty} a_m J_n(k_{n,m}x) = a_1 J_n(k_{n,1}x) + a_2 J_n(k_{n,2}x) + a_3 J_n(k_{n,3}x) + \cdots \quad (n \text{ fixed}).$$

The coefficients are (with $\alpha_{n,m} = k_{n,m}R$)

$$(10) \quad a_m = \frac{2}{R^2 J_{n+1}^2(\alpha_{n,m})} \int_0^R x f(x) J_n(k_{n,m}x) dx, \quad m = 1, 2, \dots$$

because the square of the norm is

$$(11) \quad \|J_n(k_{n,m}x)\|^2 = \int_0^R x J_n^2(k_{n,m}x) dx = \frac{R^2}{2} J_{n+1}^2(k_{n,m}R)$$

as we state without proof (which is tricky; see the discussion beginning on p. 576 of [A13]).

EXAMPLE 3 Special Fourier–Bessel Series

For instance, let us consider $f(x) = 1 - x^2$ and take $R = 1$ and $n = 0$ in the series (9), simply writing λ for $\alpha_{0,m}$. Then $k_{n,m} = \alpha_{0,m} = \lambda = 2.405, 5.520, 8.654, 11.792$, etc. (use a CAS or Table A1 in App. 5). Next we calculate the coefficients a_m by (10)

$$a_m = \frac{2}{J_1^2(\lambda)} \int_0^1 x(1 - x^2) J_0(\lambda x) dx.$$

This can be integrated by a CAS or by formulas as follows. First use $[xJ_1(\lambda x)]' = \lambda x J_0(\lambda x)$ from Theorem 1 in Sec. 5.4 and then integration by parts,

$$a_m = \frac{2}{J_1^2(\lambda)} \int_0^1 x(1 - x^2) J_0(\lambda x) dx = \frac{2}{J_1^2(\lambda)} \left[\frac{1}{\lambda} (1 - x^2) x J_1(\lambda x) \Big|_0^1 - \frac{1}{\lambda} \int_0^1 x J_1(\lambda x) (-2x) dx \right].$$

The integral-free part is zero. The remaining integral can be evaluated by $[x^2 J_2(\lambda x)]' = \lambda x^2 J_1(\lambda x)$ from Theorem 1 in Sec. 5.4. This gives

$$a_m = \frac{4J_2(\lambda)}{\lambda^2 J_1^2(\lambda)} \quad (\lambda = \alpha_{0,m}).$$

Numeric values can be obtained from a CAS (or from the table on p. 409 of Ref. [GenRef1] in App. 1, together with the formula $J_2 = 2x^{-1}J_1 - J_0$ in Theorem 1 of Sec. 5.4). This gives the eigenfunction expansion of $1 - x^2$ in terms of Bessel functions J_0 , that is,

$$1 - x^2 = 1.1081J_0(2.405x) - 0.1398J_0(5.520x) + 0.0455J_0(8.654x) - 0.0210J_0(11.792x) + \cdots$$

A graph would show that the curve of $1 - x^2$ and that of the sum of first three terms practically coincide.

Mean Square Convergence. Completeness

Ideas on approximation in the last section generalize from Fourier series to orthogonal series (1) that are made up of an orthonormal set that is “complete,” that is, consists of “sufficiently many” functions so that (1) can represent large classes of other functions (definition below).

In this connection, convergence is **convergence in the norm**, also called **mean-square convergence**; that is, a sequence of functions f_k is called **convergent with the limit** f if

$$(12^*) \quad \lim_{k \rightarrow \infty} \|f_k - f\| = 0;$$

written out by (5) in Sec. 11.5 (where we can drop the square root, as this does not affect the limit)

$$(12) \quad \lim_{k \rightarrow \infty} \int_a^b r(x)[f_k(x) - f(x)]^2 dx = 0.$$

Accordingly, the series (1) converges and represents f if

$$(13) \quad \lim_{k \rightarrow \infty} \int_a^b r(x)[s_k(x) - f(x)]^2 dx = 0$$

where s_k is the k th partial sum of (1).

$$(14) \quad s_k(x) = \sum_{m=0}^k a_m y_m(x).$$

Note that the integral in (13) generalizes (3) in Sec. 11.4.

We now define completeness. An **orthonormal** set $y_0, y_1, \dots$ on an interval $a \leq x \leq b$ is **complete** in a set of functions S defined on $a \leq x \leq b$ if we can approximate every f belonging to S arbitrarily closely in the norm by a linear combination $a_0 y_0 + a_1 y_1 + \dots + a_k y_k$, that is, technically, if for every $\epsilon > 0$ we can find constants $a_0, \dots, a_k$ (with k large enough) such that

$$(15) \quad \|f - (a_0 y_0 + \dots + a_k y_k)\| < \epsilon.$$

Ref. [GenRef7] in App. 1 uses the more modern term **total** for *complete*.

We can now extend the ideas in Sec. 11.4 that guided us from (3) in Sec. 11.4 to Bessel's and Parseval's formulas (7) and (8) in that section. Performing the square in (13) and using (14), we first have (analog of (4) in Sec. 11.4)

$$\begin{aligned} \int_a^b r(x)[s_k(x) - f(x)]^2 dx &= \int_a^b r s_k^2 dx - 2 \int_a^b r f s_k dx + \int_a^b r f^2 dx \\ &= \int_a^b r \left[\sum_{m=0}^k a_m y_m \right]^2 dx - 2 \sum_{m=0}^k a_m \int_a^b r f y_m dx + \int_a^b r f^2 dx. \end{aligned}$$

The first integral on the right equals $\sum a_m^2$ because $\int r y_m y_l dx = 0$ for $m \neq l$, and $\int r y_m^2 dx = 1$. In the second sum on the right, the integral equals a_m , by (2) with $\|y_m\|^2 = 1$. Hence the first term on the right cancels half of the second term, so that the right side reduces to (analog of (6) in Sec. 11.4)

$$- \sum_{m=0}^k a_m^2 + \int_a^b r f^2 dx.$$

This is nonnegative because in the previous formula the integrand on the left is nonnegative (recall that the weight $r(x)$ is positive!) and so is the integral on the left. This proves the important **Bessel's inequality** (analog of (7) in Sec. 11.4)

$$(16) \quad \sum_{m=0}^k a_m^2 \leq \|f\|^2 = \int_a^b r(x) f(x)^2 dx \quad (k = 1, 2, \dots),$$

Here we can let $k \rightarrow \infty$, because the left sides form a monotone increasing sequence that is bounded by the right side, so that we have convergence by the familiar Theorem 1 in App. A.3.3. Hence

$$(17) \quad \sum_{m=0}^{\infty} a_m^2 \leq \|f\|^2.$$

Furthermore, if $y_0, y_1, \dots$ is complete in a set of functions S , then (13) holds for every f belonging to S . By (13) this implies equality in (16) with $k \rightarrow \infty$. Hence in the case of completeness every f in S satisfies the so-called **Parseval equality** (analog of (8) in Sec. 11.4)

$$(18) \quad \sum_{m=0}^{\infty} a_m^2 = \|f\|^2 = \int_a^b r(x) f(x)^2 dx.$$

As a consequence of (18) we prove that in the case of *completeness* there is no function orthogonal to *every* function of the orthonormal set, with the trivial exception of a function of zero norm:

THEOREM 2

Completeness

Let $y_0, y_1, \dots$ be a complete orthonormal set on $a \leq x \leq b$ in a set of functions S . Then if a function f belongs to S and is orthogonal to every y_m , it must have norm zero. In particular, if f is continuous, then f must be identically zero.

PROOF Since f is orthogonal to every y_m , the left side of (18) must be zero. If f is continuous, then $\|f\| = 0$ implies $f(x) \equiv 0$, as can be seen directly from (5) in Sec. 11.5 with f instead of y_m because $r(x) > 0$ by assumption. ■

PROBLEM SET 11.6

1–7 FOURIER–LEGENDRE SERIES

Showing the details, develop

- $63x^5 - 90x^3 + 35x$
- $(x+1)^2$
- $1 - x^4$
- $1, x, x^2, x^3, x^4$
- Prove that if $f(x)$ is even (is odd, respectively), its Fourier–Legendre series contains only $P_m(x)$ with even m (only $P_m(x)$ with odd m , respectively). Give examples.
- What can you say about the coefficients of the Fourier–Legendre series of $f(x)$ if the Maclaurin series of $f(x)$ contains only powers x^{4m} ($m = 0, 1, 2, \dots$)?
- What happens to the Fourier–Legendre series of a polynomial $f(x)$ if you change a coefficient of $f(x)$? Experiment. Try to prove your answer.

8–13 CAS EXPERIMENT

FOURIER–LEGENDRE SERIES. Find and graph (on common axes) the partial sums up to S_{m_0} whose graph practically coincides with that of $f(x)$ within graphical accuracy. State m_0 . On what does the size of m_0 seem to depend?

- $f(x) = \sin \pi x$
- $f(x) = \sin 2\pi x$
- $f(x) = e^{-x^2}$
- $f(x) = (1 + x^2)^{-1}$
- $f(x) = J_0(\alpha_{0,1} x)$, $\alpha_{0,1}$ = the first positive zero of $J_0(x)$
- $f(x) = J_0(\alpha_{0,2} x)$, $\alpha_{0,2}$ = the second positive zero of $J_0(x)$

14. TEAM PROJECT. Orthogonality on the Entire Real Axis. Hermite Polynomials.⁸ These orthogonal polynomials are defined by $He_0(1) = 1$ and

$$(19) \quad He_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n} (e^{-x^2/2}), \quad n = 1, 2, \dots$$

REMARK. As is true for many special functions, the literature contains more than one notation, and one sometimes defines as Hermite polynomials the functions

$$H_0^* = 1, \quad H_n^*(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}.$$

This differs from our definition, which is preferred in applications.

(a) **Small Values of n .** Show that

$$\begin{aligned} He_1(x) &= x, & He_2(x) &= x^2 - 1, \\ He_3(x) &= x^3 - 3x, & He_4(x) &= x^4 - 6x^2 + 3. \end{aligned}$$

(b) **Generating Function.** A generating function of the Hermite polynomials is

$$(20) \quad e^{tx - t^2/2} = \sum_{n=0}^{\infty} a_n(x) t^n$$

because $He_n(x) = n! a_n(x)$. Prove this. *Hint:* Use the formula for the coefficients of a Maclaurin series and note that $tx - \frac{1}{2}t^2 = \frac{1}{2}x^2 - \frac{1}{2}(x - t)^2$.

(c) **Derivative.** Differentiating the generating function with respect to x , show that

$$(21) \quad He'_n(x) = n He_{n-1}(x).$$

(d) **Orthogonality on the x -Axis** needs a weight function that goes to zero sufficiently fast as $x \rightarrow \pm\infty$, (Why?)

Show that the Hermite polynomials are orthogonal on $-\infty < x < \infty$ with respect to the weight function $r(x) = e^{-x^2/2}$. *Hint.* Use integration by parts and (21).

(e) **ODEs.** Show that

$$(22) \quad He'_n(x) = x He_n(x) - He_{n+1}(x).$$

Using this with $n - 1$ instead of n and (21), show that $y = He_n(x)$ satisfies the ODE

$$(23) \quad y'' = xy' + ny = 0.$$

Show that $w = e^{-x^2/4}y$ is a solution of **Weber's equation**

$$(24) \quad w'' + (n + \frac{1}{2} - \frac{1}{4}x^2)w = 0 \quad (n = 0, 1, \dots).$$

15. CAS EXPERIMENT. Fourier-Bessel Series. Use Example 2 and $R = 1$, so that you get the series

$$(25) \quad f(x) = a_1 J_0(\alpha_{0,1}x) + a_2 J_0(\alpha_{0,2}x) + a_3 J_0(\alpha_{0,3}x) + \dots$$

With the zeros $\alpha_{0,1}\alpha_{0,2}, \dots$ from your CAS (see also Table A1 in App. 5).

(a) Graph the terms $J_0(\alpha_{0,1}x), \dots, J_0(\alpha_{0,10}x)$ for $0 \leq x \leq 1$ on common axes.

(b) Write a program for calculating partial sums of (25). Find out for what $f(x)$ your CAS can evaluate the integrals. Take two such $f(x)$ and comment empirically on the speed of convergence by observing the decrease of the coefficients.

(c) Take $f(x) = 1$ in (25) and evaluate the integrals for the coefficients analytically by (21a), Sec. 5.4, with $\nu = 1$. Graph the first few partial sums on common axes.

11.7 Fourier Integral

Fourier series are powerful tools for problems involving functions that are periodic or are of interest on a finite interval only. Sections 11.2 and 11.3 first illustrated this, and various further applications follow in Chap. 12. Since, of course, many problems involve functions that are *nonperiodic and are of interest on the whole x -axis*, we ask what can be done to extend the method of Fourier series to such functions. This idea will lead to “Fourier integrals.”

In Example 1 we start from a special function f_L of period $2L$ and see what happens to its Fourier series if we let $L \rightarrow \infty$. Then we do the same for an *arbitrary* function f_L of period $2L$. This will motivate and suggest the main result of this section, which is an integral representation given in Theorem 1 below.

⁸CHARLES HERMITE (1822–1901), French mathematician, is known for his work in algebra and number theory. The great HENRI POINCARÉ (1854–1912) was one of his students.

EXAMPLE 1 Rectangular Wave

Consider the periodic rectangular wave $f_L(x)$ of period $2L > 2$ given by

$$f_L(x) = \begin{cases} 0 & \text{if } -L < x < -1 \\ 1 & \text{if } -1 < x < 1 \\ 0 & \text{if } 1 < x < L. \end{cases}$$

The left part of Fig. 280 shows this function for $2L = 4, 8, 16$ as well as the nonperiodic function $f(x)$, which we obtain from f_L if we let $L \rightarrow \infty$,

$$f(x) = \lim_{L \rightarrow \infty} f_L(x) = \begin{cases} 1 & \text{if } -1 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

We now explore what happens to the Fourier coefficients of f_L as L increases. Since f_L is even, $b_n = 0$ for all n . For a_n the Euler formulas (6), Sec. 11.2, give

$$a_0 = \frac{1}{2L} \int_{-1}^1 dx = \frac{1}{L}, \quad a_n = \frac{1}{L} \int_{-1}^1 \cos \frac{n\pi x}{L} dx = \frac{2}{L} \int_0^1 \cos \frac{n\pi x}{L} dx = \frac{2}{L} \frac{\sin(n\pi/L)}{n\pi/L}.$$

This sequence of Fourier coefficients is called the **amplitude spectrum** of f_L because $|a_n|$ is the maximum amplitude of the wave $a_n \cos(n\pi x/L)$. Figure 280 shows this spectrum for the periods $2L = 4, 8, 16$. We see that for increasing L these amplitudes become more and more dense on the positive w_n -axis, where $w_n = n\pi/L$. Indeed, for $2L = 4, 8, 16$ we have 1, 3, 7 amplitudes per “half-wave” of the function $(2 \sin w_n)/(Lw_n)$ (dashed in the figure). Hence for $2L = 2^k$ we have $2^{k-1} - 1$ amplitudes per half-wave, so that these amplitudes will eventually be everywhere dense on the positive w_n -axis (and will decrease to zero).

The outcome of this example gives an intuitive impression of what about to expect if we turn from our special function to an arbitrary one, as we shall do next.

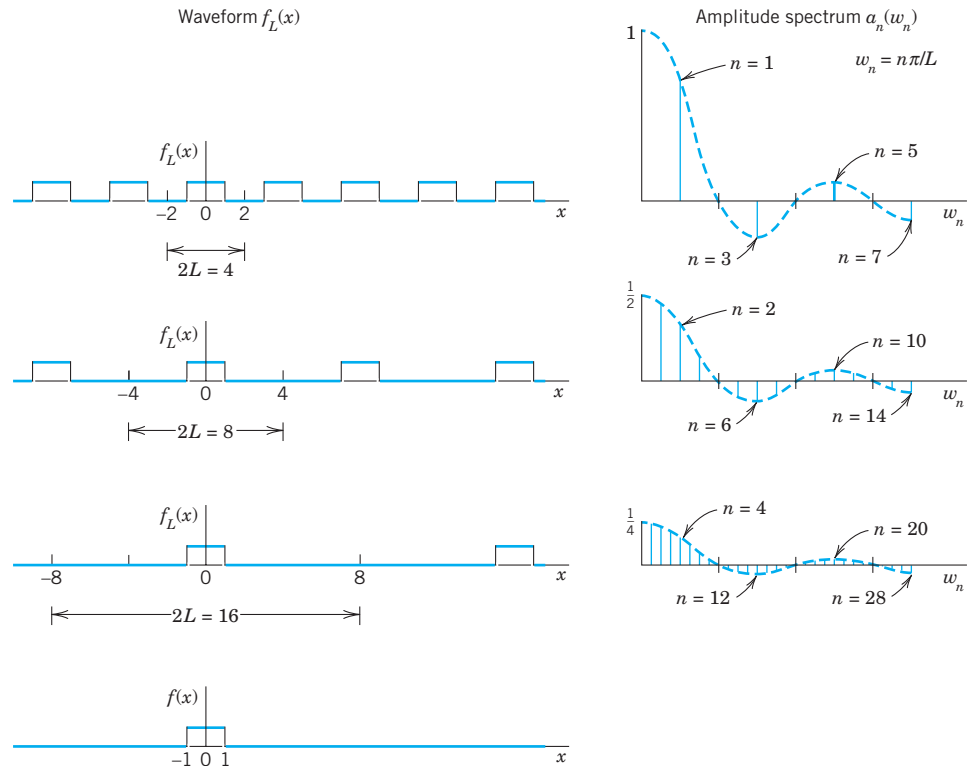


Fig. 280. Waveforms and amplitude spectra in Example 1

From Fourier Series to Fourier Integral

We now consider any periodic function $f_L(x)$ of period $2L$ that can be represented by a Fourier series

$$f_L(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos w_n x + b_n \sin w_n x), \quad w_n = \frac{n\pi}{L}$$

and find out what happens if we let $L \rightarrow \infty$. Together with Example 1 the present calculation will suggest that we should expect an integral (instead of a series) involving $\cos wx$ and $\sin wx$ with w no longer restricted to integer multiples $w = w_n = n\pi/L$ of π/L but taking *all* values. We shall also see what form such an integral might have.

If we insert a_n and b_n from the Euler formulas (6), Sec. 11.2, and denote the variable of integration by v , the Fourier series of $f_L(x)$ becomes

$$f_L(x) = \frac{1}{2L} \int_{-L}^L f_L(v) dv + \frac{1}{L} \sum_{n=1}^{\infty} \left[\cos w_n x \int_{-L}^L f_L(v) \cos w_n v dv + \sin w_n x \int_{-L}^L f_L(v) \sin w_n v dv \right].$$

We now set

$$\Delta w = w_{n+1} - w_n = \frac{(n+1)\pi}{L} - \frac{n\pi}{L} = \frac{\pi}{L}.$$

Then $1/L = \Delta w/\pi$, and we may write the Fourier series in the form

$$(1) \quad f_L(x) = \frac{1}{2L} \int_{-L}^L f_L(v) dv + \frac{1}{\pi} \sum_{n=1}^{\infty} \left[(\cos w_n x) \Delta w \int_{-L}^L f_L(v) \cos w_n v dv + (\sin w_n x) \Delta w \int_{-L}^L f_L(v) \sin w_n v dv \right].$$

This representation is valid for any fixed L , arbitrarily large, but finite.

We now let $L \rightarrow \infty$ and assume that the resulting nonperiodic function

$$f(x) = \lim_{L \rightarrow \infty} f_L(x)$$

is **absolutely integrable** on the x -axis; that is, the following (finite!) limits exist:

$$(2) \quad \lim_{a \rightarrow -\infty} \int_a^0 |f(x)| dx + \lim_{b \rightarrow \infty} \int_0^b |f(x)| dx \quad \left(\text{written } \int_{-\infty}^{\infty} |f(x)| dx \right).$$

Then $1/L \rightarrow 0$, and the value of the first term on the right side of (1) approaches zero. Also $\Delta w = \pi/L \rightarrow 0$ and it seems *plausible* that the infinite series in (1) becomes an

integral from 0 to ∞ , which represents $f(x)$, namely,

$$(3) \quad f(x) = \frac{1}{\pi} \int_0^{\infty} \left[\cos wx \int_{-\infty}^{\infty} f(v) \cos wv \, dv + \sin wx \int_{-\infty}^{\infty} f(v) \sin wv \, dv \right] dw.$$

If we introduce the notations

$$(4) \quad A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos wv \, dv, \quad B(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \sin wv \, dv$$

we can write this in the form

$$(5) \quad f(x) = \int_0^{\infty} [A(w) \cos wx + B(w) \sin wx] \, dw.$$

This is called a representation of $f(x)$ by a **Fourier integral**.

It is clear that our naive approach merely *suggests* the representation (5), but by no means establishes it; in fact, the limit of the series in (1) as Δw approaches zero is not the definition of the integral (3). Sufficient conditions for the validity of (5) are as follows.

THEOREM 1

Fourier Integral

If $f(x)$ is piecewise continuous (see Sec. 6.1) in every finite interval and has a right-hand derivative and a left-hand derivative at every point (see Sec. 11.1) and if the integral (2) exists, then $f(x)$ can be represented by a Fourier integral (5) with A and B given by (4). At a point where $f(x)$ is discontinuous the value of the Fourier integral equals the average of the left- and right-hand limits of $f(x)$ at that point (see Sec. 11.1). (Proof in Ref. [C12]; see App. 1.)

Applications of Fourier Integrals

The main application of Fourier integrals is in solving ODEs and PDEs, as we shall see for PDEs in Sec. 12.6. However, we can also use Fourier integrals in integration and in discussing functions defined by integrals, as the next example.

EXAMPLE 2

Single Pulse, Sine Integral. Dirichlet's Discontinuous Factor. Gibbs Phenomenon

Find the Fourier integral representation of the function

$$f(x) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases} \quad (\text{Fig. 281})$$

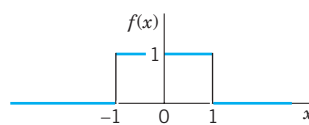


Fig. 281. Example 2

Solution. From (4) we obtain

$$A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos wv \, dv = \frac{1}{\pi} \int_{-1}^1 \cos wv \, dv = \frac{\sin wv}{\pi w} \Big|_{-1}^1 = \frac{2 \sin w}{\pi w}$$

$$B(w) = \frac{1}{\pi} \int_{-1}^1 \sin wv \, dv = 0$$

and (5) gives the *answer*

$$(6) \quad f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\cos wx \sin w}{w} dw.$$

The average of the left- and right-hand limits of $f(x)$ at $x = 1$ is equal to $(1 + 0)/2$, that is, $\frac{1}{2}$.

Furthermore, from (6) and Theorem 1 we obtain (multiply by $\pi/2$)

$$(7) \quad \int_0^{\infty} \frac{\cos wx \sin w}{w} dw = \begin{cases} \pi/2 & \text{if } 0 \leq x < 1, \\ \pi/4 & \text{if } x = 1, \\ 0 & \text{if } x > 1. \end{cases}$$

We mention that this integral is called **Dirichlet's discontinuous factor**. (For P. L. Dirichlet see Sec. 10.8.)

The case $x = 0$ is of particular interest. If $x = 0$, then (7) gives

$$(8^*) \quad \int_0^{\infty} \frac{\sin w}{w} dw = \frac{\pi}{2}.$$

We see that this integral is the limit of the so-called **sine integral**

$$(8) \quad \text{Si}(u) = \int_0^u \frac{\sin w}{w} dw$$

as $u \rightarrow \infty$. The graphs of $\text{Si}(u)$ and of the integrand are shown in Fig. 282.

In the case of a Fourier series the graphs of the partial sums are approximation curves of the curve of the periodic function represented by the series. Similarly, in the case of the Fourier integral (5), approximations are obtained by replacing ∞ by numbers a . Hence the integral

$$(9) \quad \frac{2}{\pi} \int_0^a \frac{\cos wx \sin w}{w} dw$$

approximates the right side in (6) and therefore $f(x)$.

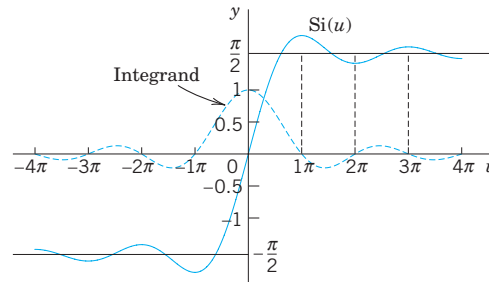


Fig. 282. Sine integral $\text{Si}(u)$ and integrand

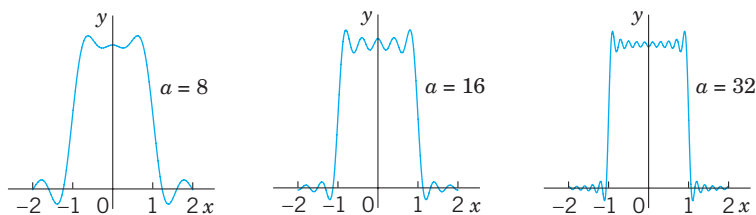


Fig. 283. The integral (9) for $a = 8, 16$, and 32 , illustrating the development of the Gibbs phenomenon

Figure 283 shows oscillations near the points of discontinuity of $f(x)$. We might expect that these oscillations disappear as a approaches infinity. But this is not true; with increasing a , they are shifted closer to the points $x = \pm 1$. This unexpected behavior, which also occurs in connection with Fourier series (see Sec. 11.2), is known as the **Gibbs phenomenon**. We can explain it by representing (9) in terms of sine integrals as follows. Using (11) in App. A3.1, we have

$$\frac{2}{\pi} \int_0^a \frac{\cos wx \sin w}{w} dw = \frac{1}{\pi} \int_0^a \frac{\sin(w + wx)}{w} dw + \frac{1}{\pi} \int_0^a \frac{\sin(w - wx)}{w} dw.$$

In the first integral on the right we set $w + wx = t$. Then $dw/w = dt/t$, and $0 \leq w \leq a$ corresponds to $0 \leq t \leq (x + 1)a$. In the last integral we set $w - wx = -t$. Then $dw/w = dt/t$, and $0 \leq w \leq a$ corresponds to $0 \leq t \leq (x - 1)a$. Since $\sin(-t) = -\sin t$, we thus obtain

$$\frac{2}{\pi} \int_0^a \frac{\cos wx \sin w}{w} dw = \frac{1}{\pi} \int_0^{(x+1)a} \frac{\sin t}{t} dt - \frac{1}{\pi} \int_0^{(x-1)a} \frac{\sin t}{t} dt.$$

From this and (8) we see that our integral (9) equals

$$\frac{1}{\pi} \text{Si}(a[x + 1]) - \frac{1}{\pi} \text{Si}(a[x - 1])$$

and the oscillations in Fig. 283 result from those in Fig. 282. The increase of a amounts to a transformation of the scale on the axis and causes the shift of the oscillations (the waves) toward the points of discontinuity -1 and 1 . ■

Fourier Cosine Integral and Fourier Sine Integral

Just as Fourier *series* simplify if a function is even or odd (see Sec. 11.2), so do Fourier *integrals*, and you can save work. Indeed, if f has a Fourier integral representation and is *even*, then $B(w) = 0$ in (4). This holds because the integrand of $B(w)$ is odd. Then (5) reduces to a **Fourier cosine integral**

$$(10) \quad f(x) = \int_0^\infty A(w) \cos wx \, dw \quad \text{where} \quad A(w) = \frac{2}{\pi} \int_0^\infty f(v) \cos wv \, dv.$$

Note the change in $A(w)$: for even f the integrand is even, hence the integral from $-\infty$ to ∞ equals twice the integral from 0 to ∞ , just as in (7a) of Sec. 11.2.

Similarly, if f has a Fourier integral representation and is *odd*, then $A(w) = 0$ in (4). This is true because the integrand of $A(w)$ is odd. Then (5) becomes a **Fourier sine integral**

$$(11) \quad f(x) = \int_0^\infty B(w) \sin wx \, dw \quad \text{where} \quad B(w) = \frac{2}{\pi} \int_0^\infty f(v) \sin wv \, dv.$$

Note the change of $B(w)$ to an integral from 0 to ∞ because $B(w)$ is even (odd times odd is even).

Earlier in this section we pointed out that the main application of the Fourier integral representation is in differential equations. However, these representations also help in evaluating integrals, as the following example shows for integrals from 0 to ∞ .

EXAMPLE 3 Laplace Integrals

We shall derive the Fourier cosine and Fourier sine integrals of $f(x) = e^{-kx}$, where $x > 0$ and $k > 0$ (Fig. 284). The result will be used to evaluate the so-called Laplace integrals.

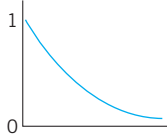


Fig. 284. $f(x)$ in Example 3

Solution. (a) From (10) we have $A(w) = \frac{2}{\pi} \int_0^{\infty} e^{-kv} \cos wv \, dv$. Now, by integration by parts,

$$\int e^{-kv} \cos wv \, dv = -\frac{k}{k^2 + w^2} e^{-kv} \left(-\frac{w}{k} \sin wv + \cos wv \right).$$

If $v = 0$, the expression on the right equals $-k/(k^2 + w^2)$. If v approaches infinity, that expression approaches zero because of the exponential factor. Thus $2/\pi$ times the integral from 0 to ∞ gives

$$(12) \quad A(w) = \frac{2k/\pi}{k^2 + w^2}.$$

By substituting this into the first integral in (10) we thus obtain the Fourier cosine integral representation

$$f(x) = e^{-kx} = \frac{2k}{\pi} \int_0^{\infty} \frac{\cos wx}{k^2 + w^2} \, dw \quad (x > 0, \quad k > 0).$$

From this representation we see that

$$(13) \quad \int_0^{\infty} \frac{\cos wx}{k^2 + w^2} \, dw = \frac{\pi}{2k} e^{-kx} \quad (x > 0, \quad k > 0).$$

(b) Similarly, from (11) we have $B(w) = \frac{2}{\pi} \int_0^{\infty} e^{-kv} \sin wv \, dv$. By integration by parts,

$$\int e^{-kv} \sin wv \, dv = -\frac{w}{k^2 + w^2} e^{-kv} \left(\frac{k}{w} \sin wv + \cos wv \right).$$

This equals $-w/(k^2 + w^2)$ if $v = 0$, and approaches 0 as $v \rightarrow \infty$. Thus

$$(14) \quad B(w) = \frac{2w/\pi}{k^2 + w^2}.$$

From (14) we thus obtain the Fourier sine integral representation

$$f(x) = e^{-kx} = \frac{2}{\pi} \int_0^{\infty} \frac{w \sin wx}{k^2 + w^2} \, dw.$$

From this we see that

$$(15) \quad \int_0^{\infty} \frac{w \sin wx}{k^2 + w^2} \, dw = \frac{\pi}{2} e^{-kx} \quad (x > 0, \quad k > 0).$$

The integrals (13) and (15) are called the **Laplace integrals**. ■

PROBLEM SET 11.7

1–6 EVALUATION OF INTEGRALS

Show that the integral represents the indicated function.
Hint. Use (5), (10), or (11); the integral tells you which one, and its value tells you what function to consider. Show your work in detail.

$$1. \int_0^{\infty} \frac{\cos xw + w \sin xw}{1 + w^2} dw = \begin{cases} 0 & \text{if } x < 0 \\ \pi/2 & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$$

$$2. \int_0^{\infty} \frac{\sin \pi w \sin xw}{1 - w^2} dw = \begin{cases} \frac{\pi}{2} \sin x & \text{if } 0 \leq x \leq \pi \\ 0 & \text{if } x > \pi \end{cases}$$

$$3. \int_0^{\infty} \frac{1 - \cos \pi w}{w} \sin xw dw = \begin{cases} \frac{1}{2}\pi & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

$$4. \int_0^{\infty} \frac{\cos \frac{1}{2} \pi w}{1 - w^2} \cos xw dw = \begin{cases} \frac{1}{2}\pi \cos x & \text{if } 0 < |x| < \frac{1}{2}\pi \\ 0 & \text{if } |x| \geq \frac{1}{2}\pi \end{cases}$$

$$5. \int_0^{\infty} \frac{\sin w - w \cos w}{w^2} \sin xw dw = \begin{cases} \frac{1}{2}\pi x & \text{if } 0 < x < 1 \\ \frac{1}{4}\pi & \text{if } x = 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$6. \int_0^{\infty} \frac{w^3 \sin xw}{w^4 + 4} dw = \frac{1}{2} \pi e^{-x} \cos x \quad \text{if } x > 0$$

7–12 FOURIER COSINE INTEGRAL REPRESENTATIONS

Represent $f(x)$ as an integral (10).

$$7. f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$8. f(x) = \begin{cases} x^2 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$9. f(x) = 1/(1 + x^2) \quad [x > 0. \text{ Hint. See (13).}]$$

$$10. f(x) = \begin{cases} a^2 - x^2 & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

$$11. f(x) = \begin{cases} \sin x & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

$$12. f(x) = \begin{cases} e^{-x} & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

13. CAS EXPERIMENT. Approximate Fourier Cosine Integrals. Graph the integrals in Prob. 7, 9, and 11 as

functions of x . Graph approximations obtained by replacing ∞ with finite upper limits of your choice. Compare the quality of the approximations. Write a short report on your empirical results and observations.

14. PROJECT. Properties of Fourier Integrals

(a) **Fourier cosine integral.** Show that (10) implies

$$(a1) \quad f(ax) = \frac{1}{a} \int_0^{\infty} A\left(\frac{w}{a}\right) \cos xw dw$$

($a > 0$) (Scale change)

$$(a2) \quad xf(x) = \int_0^{\infty} B^*(w) \sin xw dw,$$

$$B^* = -\frac{dA}{dw}, \quad A \text{ as in (10)}$$

$$(a3) \quad x^2 f(x) = \int_0^{\infty} A^*(w) \cos xw dw,$$

$$A^* = -\frac{d^2 A}{dw^2}.$$

(b) Solve Prob. 8 by applying (a3) to the result of Prob. 7.

(c) Verify (a2) for $f(x) = 1$ if $0 < x < a$ and $f(x) = 0$ if $x > a$.

(d) **Fourier sine integral.** Find formulas for the Fourier sine integral similar to those in (a).

15. CAS EXPERIMENT. Sine Integral. Plot $\text{Si}(u)$ for positive u . Does the sequence of the maximum and minimum values give the impression that it converges and has the limit $\pi/2$? Investigate the Gibbs phenomenon graphically.

16–20 FOURIER SINE INTEGRAL REPRESENTATIONS

Represent $f(x)$ as an integral (11).

$$16. f(x) = \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$$

$$17. f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$18. f(x) = \begin{cases} \cos x & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

$$19. f(x) = \begin{cases} e^x & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

$$20. f(x) = \begin{cases} e^{-x} & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$$

11.8 Fourier Cosine and Sine Transforms

An **integral transform** is a transformation in the form of an integral that produces from given functions new functions depending on a different variable. One is mainly interested in these transforms because they can be used as tools in solving ODEs, PDEs, and integral equations and can often be of help in handling and applying special functions. The Laplace transform of Chap. 6 serves as an example and is by far the most important integral transform in engineering.

Next in order of importance are Fourier transforms. They can be obtained from the Fourier integral in Sec. 11.7 in a straightforward way. In this section we derive two such transforms that are real, and in Sec. 11.9 a complex one.

Fourier Cosine Transform

The Fourier cosine transform concerns **even functions** $f(x)$. We obtain it from the Fourier cosine integral [(10) in Sec. 10.7]

$$f(x) = \int_0^{\infty} A(w) \cos wx \, dw, \quad \text{where} \quad A(w) = \frac{2}{\pi} \int_0^{\infty} f(v) \cos wv \, dv.$$

Namely, we set $A(w) = \sqrt{2/\pi} \hat{f}_c(w)$, where c suggests “cosine.” Then, writing $v = x$ in the formula for $A(w)$, we have

$$(1a) \quad \hat{f}_c(w) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos wx \, dx$$

and

$$(1b) \quad f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \hat{f}_c(w) \cos wx \, dw.$$

Formula (1a) gives from $f(x)$ a new function $\hat{f}_c(w)$, called the **Fourier cosine transform** of $f(x)$. Formula (1b) gives us back $f(x)$ from $\hat{f}_c(w)$, and we therefore call $f(x)$ the **inverse Fourier cosine transform** of $\hat{f}_c(w)$.

The process of obtaining the transform $\hat{f}_c$ from a given f is also called the **Fourier cosine transform** or the *Fourier cosine transform method*.

Fourier Sine Transform

Similarly, in (11), Sec. 11.7, we set $B(w) = \sqrt{2/\pi} \hat{f}_s(w)$, where s suggests “sine.” Then, writing $v = x$, we have from (11), Sec. 11.7, the **Fourier sine transform**, of $f(x)$ given by

$$(2a) \quad \hat{f}_s(w) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin wx \, dx,$$

and the **inverse Fourier sine transform** of $\hat{f}_s(w)$, given by

$$(2b) \quad f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \hat{f}_s(w) \sin wx \, dw.$$

The process of obtaining $f_s(w)$ from $f(x)$ is also called the **Fourier sine transform** or the *Fourier sine transform method*.

Other notations are

$$\mathcal{F}_c(f) = \hat{f}_c, \quad \mathcal{F}_s(f) = \hat{f}_s$$

and $\mathcal{F}_c^{-1}$ and $\mathcal{F}_s^{-1}$ for the inverses of $\mathcal{F}_c$ and $\mathcal{F}_s$, respectively.

EXAMPLE 1 Fourier Cosine and Fourier Sine Transforms

Find the Fourier cosine and Fourier sine transforms of the function

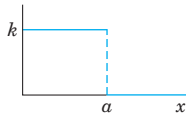


Fig. 285. $f(x)$ in Example 1

$$f(x) = \begin{cases} k & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases} \quad (\text{Fig. 285}).$$

Solution. From the definitions (1a) and (2a) we obtain by integration

$$\begin{aligned} \hat{f}_c(w) &= \sqrt{\frac{2}{\pi}} k \int_0^a \cos wx \, dx = \sqrt{\frac{2}{\pi}} k \left(\frac{\sin aw}{w} \right) \\ \hat{f}_s(w) &= \sqrt{\frac{2}{\pi}} k \int_0^a \sin wx \, dx = \sqrt{\frac{2}{\pi}} k \left(\frac{1 - \cos aw}{w} \right). \end{aligned}$$

This agrees with formulas 1 in the first two tables in Sec. 11.10 (where $k = 1$).

Note that for $f(x) = k = \text{const}$ ($0 < x < \infty$), these transforms do not exist. (Why?)

EXAMPLE 2 Fourier Cosine Transform of the Exponential Function

Find $\mathcal{F}_c(e^{-x})$.

Solution. By integration by parts and recursion,

$$\mathcal{F}_c(e^{-x}) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-x} \cos wx \, dx = \sqrt{\frac{2}{\pi}} \frac{e^{-x}}{1 + w^2} (-\cos wx + w \sin wx) \Big|_0^{\infty} = \frac{\sqrt{2/\pi}}{1 + w^2}.$$

This agrees with formula 3 in Table I, Sec. 11.10, with $a = 1$. See also the next example.

What did we do to introduce the two integral transforms under consideration? Actually not much: We changed the notations A and B to get a “symmetric” distribution of the constant $2/\pi$ in the original formulas (1) and (2). This redistribution is a standard convenience, but it is not essential. One could do without it.

What have we gained? We show next that these transforms have operational properties that permit them to convert differentiations into algebraic operations (just as the Laplace transform does). This is the key to their application in solving differential equations.

Linearity, Transforms of Derivatives

If $f(x)$ is absolutely integrable (see Sec. 11.7) on the positive x -axis and piecewise continuous (see Sec. 6.1) on every finite interval, then the Fourier cosine and sine transforms of f exist.

Furthermore, if f and g have Fourier cosine and sine transforms, so does $af + bg$ for any constants a and b , and by (1a)

$$\begin{aligned}\mathcal{F}_c(af + bg) &= \sqrt{\frac{2}{\pi}} \int_0^\infty [af(x) + bg(x)] \cos wx \, dx \\ &= a \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos wx \, dx + b \sqrt{\frac{2}{\pi}} \int_0^\infty g(x) \cos wx \, dx.\end{aligned}$$

The right side is $a\mathcal{F}_c(f) + b\mathcal{F}_c(g)$. Similarly for $\mathcal{F}_s$, by (2). This shows that the Fourier cosine and sine transforms are **linear operations**,

$$\begin{aligned}(3) \quad (a) \quad &\mathcal{F}_c(af + bg) = a\mathcal{F}_c(f) + b\mathcal{F}_c(g), \\ (b) \quad &\mathcal{F}_s(af + bg) = a\mathcal{F}_s(f) + b\mathcal{F}_s(g).\end{aligned}$$

THEOREM 1

Cosine and Sine Transforms of Derivatives

Let $f(x)$ be continuous and absolutely integrable on the x -axis, let $f'(x)$ be piecewise continuous on every finite interval, and let $f(x) \rightarrow 0$ as $x \rightarrow \infty$. Then

$$\begin{aligned}(4) \quad (a) \quad &\mathcal{F}_c\{f'(x)\} = w\mathcal{F}_s\{f(x)\} - \sqrt{\frac{2}{\pi}}f(0), \\ (b) \quad &\mathcal{F}_s\{f'(x)\} = -w\mathcal{F}_c\{f(x)\}.\end{aligned}$$

PROOF This follows from the definitions and by using integration by parts, namely,

$$\begin{aligned}\mathcal{F}_c\{f'(x)\} &= \sqrt{\frac{2}{\pi}} \int_0^\infty f'(x) \cos wx \, dx \\ &= \sqrt{\frac{2}{\pi}} \left[f(x) \cos wx \Big|_0^\infty + w \int_0^\infty f(x) \sin wx \, dx \right] \\ &= -\sqrt{\frac{2}{\pi}} f(0) + w\mathcal{F}_s\{f(x)\};\end{aligned}$$

and similarly,

$$\begin{aligned}\mathcal{F}_s\{f'(x)\} &= \sqrt{\frac{2}{\pi}} \int_0^\infty f'(x) \sin wx \, dx \\ &= \sqrt{\frac{2}{\pi}} \left[f(x) \sin wx \Big|_0^\infty - w \int_0^\infty f(x) \cos wx \, dx \right] \\ &= 0 - w\mathcal{F}_c\{f(x)\}.\end{aligned}$$

Formula (4a) with f' instead of f gives (when f', f'' satisfy the respective assumptions for f, f' in Theorem 1)

$$\mathcal{F}_c\{f''(x)\} = w\mathcal{F}_s\{f'(x)\} - \sqrt{\frac{2}{\pi}}f'(0);$$

hence by (4b)

$$(5a) \quad \mathcal{F}_c\{f''(x)\} = -w^2\mathcal{F}_c\{f(x)\} - \sqrt{\frac{2}{\pi}}f'(0).$$

Similarly,

$$(5b) \quad \mathcal{F}_s\{f''(x)\} = -w^2\mathcal{F}_s\{f(x)\} + \sqrt{\frac{2}{\pi}}wf(0).$$

A basic application of (5) to PDEs will be given in Sec. 12.7. For the time being we show how (5) can be used for deriving transforms.

EXAMPLE 3 An Application of the Operational Formula (5)

Find the Fourier cosine transform $\mathcal{F}_c(e^{-ax})$ of $f(x) = e^{-ax}$, where $a > 0$.

Solution. By differentiation, $(e^{-ax})'' = a^2e^{-ax}$; thus

$$a^2f(x) = f''(x).$$

From this, (5a), and the linearity (3a),

$$\begin{aligned} a^2\mathcal{F}_c(f) &= \mathcal{F}_c(f'') \\ &= -w^2\mathcal{F}_c(f) - \sqrt{\frac{2}{\pi}}f'(0) \\ &= -w^2\mathcal{F}_c(f) + a\sqrt{\frac{2}{\pi}}. \end{aligned}$$

Hence

$$(a^2 + w^2)\mathcal{F}_c(f) = a\sqrt{2/\pi}.$$

The *answer* is (see Table I, Sec. 11.10)

$$\mathcal{F}_c(e^{-ax}) = \sqrt{\frac{2}{\pi}}\left(\frac{a}{a^2 + w^2}\right) \quad (a > 0). \quad \blacksquare$$

Tables of Fourier cosine and sine transforms are included in Sec. 11.10.

PROBLEM SET 11.8

1–8 FOURIER COSINE TRANSFORM

1. Find the cosine transform $\hat{f}_c(w)$ of $f(x) = 1$ if $0 < x < 1$, $f(x) = -1$ if $1 < x < 2$, $f(x) = 0$ if $x > 2$.
2. Find f in Prob. 1 from the answer $\hat{f}_c$.
3. Find $\hat{f}_c(w)$ for $f(x) = x$ if $0 < x < 2$, $f(x) = 0$ if $x > 2$.
4. Derive formula 3 in Table I of Sec. 11.10 by integration.
5. Find $\hat{f}_c(w)$ for $f(x) = x^2$ if $0 < x < 1$, $f(x) = 0$ if $x > 1$.
6. **Continuity assumptions.** Find $\hat{g}_c(w)$ for $g(x) = 2$ if $0 < x < 1$, $g(x) = 0$ if $x > 1$. Try to obtain from it $\hat{f}_c(w)$ for $f(x)$ in Prob. 5 by using (5a).
7. **Existence?** Does the Fourier cosine transform of $x^{-1} \sin x$ ($0 < x < \infty$) exist? Of $x^{-1} \cos x$? Give reasons.
8. **Existence?** Does the Fourier cosine transform of $f(x) = k = \text{const}$ ($0 < x < \infty$) exist? The Fourier sine transform?

9–15 FOURIER SINE TRANSFORM

9. Find $\mathcal{F}_s(e^{-ax})$, $a > 0$, by integration.
10. Obtain the answer to Prob. 9 from (5b).
11. Find $f_s(w)$ for $f(x) = x^2$ if $0 < x < 1$, $f(x) = 0$ if $x > 1$.
12. Find $\mathcal{F}_s(xe^{-x^2/2})$ from (4b) and a suitable formula in Table I of Sec. 11.10.
13. Find $\mathcal{F}_s(e^{-x})$ from (4a) and formula 3 of Table I in Sec. 11.10.
14. **Gamma function.** Using formulas 2 and 4 in Table II of Sec. 11.10, prove $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ [(30) in App. A3.1], a value needed for Bessel functions and other applications.
15. **WRITING PROJECT. Finding Fourier Cosine and Sine Transforms.** Write a short report on ways of obtaining these transforms, with illustrations by examples of your own.

11.9 Fourier Transform. Discrete and Fast Fourier Transforms

In Sec. 11.8 we derived two real transforms. Now we want to derive a complex transform that is called the **Fourier transform**. It will be obtained from the complex Fourier integral, which will be discussed next.

Complex Form of the Fourier Integral

The (real) Fourier integral is [see (4), (5), Sec. 11.7]

$$f(x) = \int_0^{\infty} [A(w) \cos wx + B(w) \sin wx] dw$$

where

$$A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos wv dv, \quad B(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \sin wv dv.$$

Substituting A and B into the integral for f , we have

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(v) [\cos wv \cos wx + \sin wv \sin wx] dv dw.$$

By the addition formula for the cosine [(6) in App. A3.1] the expression in the brackets $[\cdots]$ equals $\cos(wv - wx)$ or, since the cosine is even, $\cos(wx - wv)$. We thus obtain

$$(1^*) \quad f(x) = \frac{1}{\pi} \int_0^\infty \left[\int_{-\infty}^\infty f(v) \cos(wx - wv) dv \right] dw.$$

The integral in brackets is an *even* function of w , call it $F(w)$, because $\cos(wx - wv)$ is an even function of w , the function f does not depend on w , and we integrate with respect to v (not w). Hence the integral of $F(w)$ from $w = 0$ to ∞ is $\frac{1}{2}$ times the integral of $F(w)$ from $-\infty$ to ∞ . Thus (note the change of the integration limit!)

$$(1) \quad f(x) = \frac{1}{2\pi} \int_{-\infty}^\infty \left[\int_{-\infty}^\infty f(v) \cos(wx - wv) dv \right] dw.$$

We claim that the integral of the form (1) with $\sin$ instead of $\cos$ is zero:

$$(2) \quad \frac{1}{2\pi} \int_{-\infty}^\infty \left[\int_{-\infty}^\infty f(v) \sin(wx - wv) dv \right] dw = 0.$$

This is true since $\sin(wx - wv)$ is an odd function of w , which makes the integral in brackets an odd function of w , call it $G(w)$. Hence the integral of $G(w)$ from $-\infty$ to ∞ is zero, as claimed.

We now take the integrand of (1) plus $i (= \sqrt{-1})$ times the integrand of (2) and use the **Euler formula** [(11) in Sec. 2.2]

$$(3) \quad e^{ix} = \cos x + i \sin x.$$

Taking $wv - wx$ instead of x in (3) and multiplying by $f(v)$ gives

$$f(v) \cos(wx - wv) + if(v) \sin(wx - wv) = f(v) e^{i(wx - wv)}.$$

Hence the result of adding (1) plus i times (2), called the **complex Fourier integral**, is

$$(4) \quad f(x) = \frac{1}{2\pi} \int_{-\infty}^\infty \int_{-\infty}^\infty f(v) e^{iwx - wv} dv dw \quad (i = \sqrt{-1}).$$

To obtain the desired Fourier transform will take only a very short step from here.

Fourier Transform and Its Inverse

Writing the exponential function in (4) as a product of exponential functions, we have

$$(5) \quad f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty f(v) e^{-iwx} dv \right] e^{iwx} dw.$$

The expression in brackets is a function of w , is denoted by $\hat{f}(w)$, and is called the **Fourier transform** of f ; writing $v = x$, we have

$$(6) \quad \hat{f}(w) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty f(x) e^{-iwx} dx.$$

With this, (5) becomes

$$(7) \quad f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(w) e^{iwx} dw$$

and is called the **inverse Fourier transform** of $\hat{f}(w)$.

Another notation for the Fourier transform is

$$\hat{f} = \mathcal{F}(f),$$

so that

$$f = \mathcal{F}^{-1}(\hat{f}).$$

The process of obtaining the Fourier transform $\mathcal{F}(f) = \hat{f}$ from a given f is also called the **Fourier transform** or the *Fourier transform method*.

Using concepts defined in Secs. 6.1 and 11.7 we now state (without proof) conditions that are sufficient for the existence of the Fourier transform.

THEOREM 1

Existence of the Fourier Transform

If $f(x)$ is absolutely integrable on the x -axis and piecewise continuous on every finite interval, then the Fourier transform $\hat{f}(w)$ of $f(x)$ given by (6) exists.

EXAMPLE 1

Fourier Transform

Find the Fourier transform of $f(x) = 1$ if $|x| < 1$ and $f(x) = 0$ otherwise.

Solution. Using (6) and integrating, we obtain

$$\hat{f}(w) = \frac{1}{\sqrt{2\pi}} \int_{-1}^1 e^{-iwx} dx = \frac{1}{\sqrt{2\pi}} \cdot \left. \frac{e^{-iwx}}{-iw} \right|_{-1}^1 = \frac{1}{-iw\sqrt{2\pi}} (e^{-iw} - e^{iw}).$$

As in (3) we have $e^{iw} = \cos w + i \sin w$, $e^{-iw} = \cos w - i \sin w$, and by subtraction

$$e^{iw} - e^{-iw} = 2i \sin w.$$

Substituting this in the previous formula on the right, we see that i drops out and we obtain the answer

$$\hat{f}(w) = \sqrt{\frac{\pi}{2}} \frac{\sin w}{w}.$$

EXAMPLE 2

Fourier Transform

Find the Fourier transform $\mathcal{F}(e^{-ax})$ of $f(x) = e^{-ax}$ if $x > 0$ and $f(x) = 0$ if $x < 0$; here $a > 0$.

Solution. From the definition (6) we obtain by integration

$$\begin{aligned} \mathcal{F}(e^{-ax}) &= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-ax} e^{-iwx} dx \\ &= \frac{1}{\sqrt{2\pi}} \left. \frac{e^{-(a+iw)x}}{-(a+iw)} \right|_{x=0}^{\infty} = \frac{1}{\sqrt{2\pi}(a+iw)}. \end{aligned}$$

This proves formula 5 of Table III in Sec. 11.10.

Physical Interpretation: Spectrum

The nature of the representation (7) of $f(x)$ becomes clear if we think of it as a superposition of sinusoidal oscillations of all possible frequencies, called a **spectral representation**. This name is suggested by optics, where light is such a superposition of colors (frequencies). In (7), the “**spectral density**” $\hat{f}(w)$ measures the intensity of $f(x)$ in the frequency interval between w and $w + \Delta w$ (Δw small, fixed). We claim that, in connection with vibrations, the integral

$$\int_{-\infty}^{\infty} |\hat{f}(w)|^2 dw$$

can be interpreted as the **total energy** of the physical system. Hence an integral of $|\hat{f}(w)|^2$ from a to b gives the contribution of the frequencies w between a and b to the total energy.

To make this plausible, we begin with a mechanical system giving a single frequency, namely, the harmonic oscillator (mass on a spring, Sec. 2.4)

$$my'' + ky = 0.$$

Here we denote time t by x . Multiplication by y' gives $my'y'' + ky'y = 0$. By integration,

$$\frac{1}{2}mv^2 + \frac{1}{2}ky^2 = E_0 = \text{const}$$

where $v = y'$ is the velocity. The first term is the kinetic energy, the second the potential energy, and E_0 the total energy of the system. Now a general solution is (use (3) in Sec. 11.4 with $t = x$)

$$y = a_1 \cos w_0 x + b_1 \sin w_0 x = c_1 e^{iw_0 x} + c_{-1} e^{-iw_0 x}, \quad w_0^2 = k/m$$

where $c_1 = (a_1 - ib_1)/2$, $c_{-1} = \bar{c}_1 = (a_1 + ib_1)/2$. We write simply $A = c_1 e^{iw_0 x}$, $B = c_{-1} e^{-iw_0 x}$. Then $y = A + B$. By differentiation, $v = y' = A' + B' = iw_0(A - B)$. Substitution of v and y on the left side of the equation for E_0 gives

$$E_0 = \frac{1}{2}mv^2 + \frac{1}{2}ky^2 = \frac{1}{2}m(iw_0)^2(A - B)^2 + \frac{1}{2}k(A + B)^2.$$

Here $w_0^2 = k/m$, as just stated; hence $mw_0^2 = k$. Also $i^2 = -1$, so that

$$E_0 = \frac{1}{2}k[-(A - B)^2 + (A + B)^2] = 2kAB = 2kc_1 e^{iw_0 x} c_{-1} e^{-iw_0 x} = 2kc_1 c_{-1} = 2k|c_1|^2.$$

Hence *the energy is proportional to the square of the amplitude* $|c_1|$.

As the next step, if a more complicated system leads to a periodic solution $y = f(x)$ that can be represented by a Fourier series, then instead of the single energy term $|c_1|^2$ we get a series of squares $|c_n|^2$ of Fourier coefficients c_n given by (6), Sec. 11.4. In this case we have a “**discrete spectrum**” (or “**point spectrum**”) consisting of countably many isolated frequencies (infinitely many, in general), the corresponding $|c_n|^2$ being the contributions to the total energy.

Finally, a system whose solution can be represented by an integral (7) leads to the above integral for the energy, as is plausible from the cases just discussed.

Linearity. Fourier Transform of Derivatives

New transforms can be obtained from given ones by using

THEOREM 2

Linearity of the Fourier Transform

The Fourier transform is a **linear operation**; that is, for any functions $f(x)$ and $g(x)$ whose Fourier transforms exist and any constants a and b , the Fourier transform of $af + bg$ exists, and

$$(8) \quad \mathcal{F}(af + bg) = a\mathcal{F}(f) + b\mathcal{F}(g).$$

PROOF This is true because integration is a linear operation, so that (6) gives

$$\begin{aligned} \mathcal{F}\{af(x) + bg(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} [af(x) + bg(x)] e^{-iwx} dx \\ &= a \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} dx + b \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x) e^{-iwx} dx \\ &= a\mathcal{F}\{f(x)\} + b\mathcal{F}\{g(x)\}. \end{aligned}$$

In applying the Fourier transform to differential equations, the key property is that differentiation of functions corresponds to multiplication of transforms by iw :

THEOREM 3

Fourier Transform of the Derivative of $f(x)$

Let $f(x)$ be continuous on the x -axis and $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$. Furthermore, let $f'(x)$ be absolutely integrable on the x -axis. Then

$$(9) \quad \mathcal{F}\{f'(x)\} = iw\mathcal{F}\{f(x)\}.$$

PROOF From the definition of the Fourier transform we have

$$\mathcal{F}\{f'(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f'(x) e^{-iwx} dx.$$

Integrating by parts, we obtain

$$\mathcal{F}\{f'(x)\} = \frac{1}{\sqrt{2\pi}} \left[f(x) e^{-iwx} \Big|_{-\infty}^{\infty} - (-iw) \int_{-\infty}^{\infty} f(x) e^{-iwx} dx \right].$$

Since $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$, the desired result follows, namely,

$$\mathcal{F}\{f'(x)\} = 0 + iw\mathcal{F}\{f(x)\}.$$

Two successive applications of (9) give

$$\mathcal{F}(f'') = iw\mathcal{F}(f') = (iw)^2\mathcal{F}(f).$$

Since $(iw)^2 = -w^2$, we have for the transform of the second derivative of f

$$(10) \quad \mathcal{F}\{f''(x)\} = -w^2\mathcal{F}\{f(x)\}.$$

Similarly for higher derivatives.

An application of (10) to differential equations will be given in Sec. 12.6. For the time being we show how (9) can be used to derive transforms.

EXAMPLE 3 Application of the Operational Formula (9)

Find the Fourier transform of xe^{-x^2} from Table III, Sec 11.10.

Solution. We use (9). By formula 9 in Table III

$$\begin{aligned} \mathcal{F}(xe^{-x^2}) &= \mathcal{F}\left\{-\frac{1}{2}(e^{-x^2})'\right\} \\ &= -\frac{1}{2}\mathcal{F}\{(e^{-x^2})'\} \\ &= -\frac{1}{2}iw\mathcal{F}(e^{-x^2}) \\ &= -\frac{1}{2}iw\frac{1}{\sqrt{2}}e^{-w^2/4} \\ &= -\frac{iw}{2\sqrt{2}}e^{-w^2/4}. \end{aligned}$$

Convolution

The **convolution** $f * g$ of functions f and g is defined by

$$(11) \quad h(x) = (f * g)(x) = \int_{-\infty}^{\infty} f(p)g(x-p)dp = \int_{-\infty}^{\infty} f(x-p)g(p)dp.$$

The purpose is the same as in the case of Laplace transforms (Sec. 6.5): taking the convolution of two functions and then taking the transform of the convolution is the same as multiplying the transforms of these functions (and multiplying them by $\sqrt{2\pi}$):

THEOREM 4

Convolution Theorem

Suppose that $f(x)$ and $g(x)$ are piecewise continuous, bounded, and absolutely integrable on the x -axis. Then

$$(12) \quad \mathcal{F}(f * g) = \sqrt{2\pi} \mathcal{F}(f) \mathcal{F}(g).$$

PROOF By the definition,

$$\mathcal{F}(f * g) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p) g(x - p) dp e^{-iwx} dx.$$

An interchange of the order of integration gives

$$\mathcal{F}(f * g) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p) g(x - p) e^{-iwx} dx dp.$$

Instead of x we now take $x - p = q$ as a new variable of integration. Then $x = p + q$ and

$$\mathcal{F}(f * g) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p) g(q) e^{-iwp(p+q)} dq dp.$$

This double integral can be written as a product of two integrals and gives the desired result

$$\begin{aligned} \mathcal{F}(f * g) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(p) e^{-iwp} dp \int_{-\infty}^{\infty} g(q) e^{-iwp} dq \\ &= \frac{1}{\sqrt{2\pi}} [\sqrt{2\pi} \mathcal{F}(f)] [\sqrt{2\pi} \mathcal{F}(g)] = \sqrt{2\pi} \mathcal{F}(f) \mathcal{F}(g). \end{aligned}$$

By taking the inverse Fourier transform on both sides of (12), writing $\hat{f} = \mathcal{F}(f)$ and $\hat{g} = \mathcal{F}(g)$ as before, and noting that $\sqrt{2\pi}$ and $1/\sqrt{2\pi}$ in (12) and (7) cancel each other, we obtain

$$(13) \quad (f * g)(x) = \int_{-\infty}^{\infty} \hat{f}(w) \hat{g}(w) e^{iwx} dw,$$

a formula that will help us in solving partial differential equations (Sec. 12.6).

Discrete Fourier Transform (DFT), Fast Fourier Transform (FFT)

In using Fourier series, Fourier transforms, and trigonometric approximations (Sec. 11.6) we have to assume that a function $f(x)$, to be developed or transformed, is given on some interval, over which we integrate in the Euler formulas, etc. Now very often a function $f(x)$ is given only in terms of values at finitely many points, and one is interested in extending Fourier analysis to this case. The main application of such a “discrete Fourier analysis” concerns large amounts of equally spaced data, as they occur in telecommunication, time series analysis, and various simulation problems. In these situations, dealing with sampled values rather than with functions, we can replace the Fourier transform by the so-called **discrete Fourier transform (DFT)** as follows.

Let $f(x)$ be periodic, for simplicity of period 2π . We assume that N measurements of $f(x)$ are taken over the interval $0 \leq x \leq 2\pi$ at regularly spaced points

$$(14) \quad x_k = \frac{2\pi k}{N}, \quad k = 0, 1, \dots, N-1.$$

We also say that $f(x)$ is being **sampled** at these points. We now want to determine a **complex trigonometric polynomial**

$$(15) \quad q(x) = \sum_{n=0}^{N-1} c_n e^{inx_k}$$

that **interpolates** $f(x)$ at the nodes (14), that is, $q(x_k) = f(x_k)$, written out, with f_k denoting $f(x_k)$,

$$(16) \quad f_k = f(x_k) = q(x_k) = \sum_{n=0}^{N-1} c_n e^{inx_k}, \quad k = 0, 1, \dots, N-1.$$

Hence we must determine the coefficients $c_0, \dots, c_{N-1}$ such that (16) holds. We do this by an idea similar to that in Sec. 11.1 for deriving the Fourier coefficients by using the orthogonality of the trigonometric system. Instead of integrals we now take sums. Namely, we multiply (16) by e^{-imx_k} (note the minus!) and sum over k from 0 to $N-1$. Then we interchange the order of the two summations and insert x_k from (14). This gives

$$(17) \quad \sum_{k=0}^{N-1} f_k e^{-imx_k} = \sum_{k=0}^{N-1} \sum_{n=0}^{N-1} c_n e^{i(n-m)x_k} = \sum_{n=0}^{N-1} c_n \sum_{k=0}^{N-1} e^{i(n-m)2\pi k/N}.$$

Now

$$e^{i(n-m)2\pi k/N} = [e^{i(n-m)2\pi/N}]^k.$$

We denote $[\dots]$ by r . For $n = m$ we have $r = e^0 = 1$. The sum of *these* terms over k equals N , the number of these terms. For $n \neq m$ we have $r \neq 1$ and by the formula for a geometric sum [(6) in Sec. 15.1 with $q = r$ and $n = N-1$]

$$\sum_{k=0}^{N-1} r^k = \frac{1 - r^N}{1 - r} = 0$$

because $r^N = 1$; indeed, since k, m , and n are integers,

$$r^N = e^{i(n-m)2\pi} = \cos 2\pi k(n-m) + i \sin 2\pi k(n-m) = 1 + 0 = 1.$$

This shows that the right side of (17) equals $c_m N$. Writing n for m and dividing by N , we thus obtain the desired coefficient formula

$$(18^*) \quad c_n = \frac{1}{N} \sum_{k=0}^{N-1} f_k e^{-inx_k} \quad f_k = f(x_k), \quad n = 0, 1, \dots, N-1.$$

Since computation of the c_n (by the fast Fourier transform, below) involves successive halving of the problem size N , it is practical to drop the factor $1/N$ from c_n and define the

discrete Fourier transform of the given signal $\mathbf{f} = [f_0 \ \cdots \ f_{N-1}]^T$ to be the vector $\hat{\mathbf{f}} = [\hat{f}_0 \ \cdots \ \hat{f}_{N-1}]$ with components

$$(18) \quad \hat{f}_n = Nc_n = \sum_{k=0}^{N-1} f_k e^{-inx_k}, \quad f_k = f(x_k), \quad n = 0, \dots, N-1.$$

This is the frequency spectrum of the signal.

In vector notation, $\hat{\mathbf{f}} = \mathbf{F}_N \mathbf{f}$, where the $N \times N$ **Fourier matrix** $\mathbf{F}_N = [e_{nk}]$ has the entries [given in (18)]

$$(19) \quad e_{nk} = e^{-inx_k} = e^{-2\pi ink/N} = w^{nk}, \quad w = w_N = e^{-2\pi i/N},$$

where $n, k = 0, \dots, N-1$.

EXAMPLE 4 Discrete Fourier Transform (DFT). Sample of $N = 4$ Values

Let $N = 4$ measurements (sample values) be given. Then $w = e^{-2\pi i/N} = e^{-\pi i/2} = -i$ and thus $w^{nk} = (-i)^{nk}$. Let the sample values be, say $\mathbf{f} = [0 \ 1 \ 4 \ 9]^T$. Then by (18) and (19),

$$(20) \quad \hat{\mathbf{f}} = \mathbf{F}_4 \mathbf{f} = \begin{bmatrix} w^0 & w^0 & w^0 & w^0 \\ w^0 & w^1 & w^2 & w^3 \\ w^0 & w^2 & w^4 & w^6 \\ w^0 & w^3 & w^6 & w^9 \end{bmatrix} \mathbf{f} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 4 \\ 9 \end{bmatrix} = \begin{bmatrix} 14 \\ -4 + 8i \\ -6 \\ -4 - 8i \end{bmatrix}.$$

From the first matrix in (20) it is easy to infer what $\mathbf{F}_N$ looks like for arbitrary N , which in practice may be 1000 or more, for reasons given below. ■

From the DFT (the frequency spectrum) $\hat{\mathbf{f}} = \mathbf{F}_N \mathbf{f}$ we can recreate the given signal $\hat{\mathbf{f}} = \mathbf{F}_N^{-1} \mathbf{f}$, as we shall now prove. Here $\mathbf{F}_N$ and its complex conjugate $\bar{\mathbf{F}}_N = \frac{1}{N} [\bar{w}^{nk}]$ satisfy

$$(21a) \quad \bar{\mathbf{F}}_N \mathbf{F}_N = \mathbf{F}_N \bar{\mathbf{F}}_N = N \mathbf{I}$$

where $\mathbf{I}$ is the $N \times N$ unit matrix; hence $\mathbf{F}_N$ has the inverse

$$(21b) \quad \mathbf{F}_N^{-1} = \frac{1}{N} \bar{\mathbf{F}}_N.$$

PROOF We prove (21). By the multiplication rule (row times column) the product matrix $\mathbf{G}_N = \bar{\mathbf{F}}_N \mathbf{F}_N = [g_{jk}]$ in (21a) has the entries $g_{jk} = \text{Row } j \text{ of } \bar{\mathbf{F}}_N \text{ times Column } k \text{ of } \mathbf{F}_N$. That is, writing $W = \bar{w}^j w^k$, we prove that

$$\begin{aligned} g_{jk} &= (\bar{w}^j w^k)^0 + (\bar{w}^j w^k)^1 + \cdots + (\bar{w}^j w^k)^{N-1} \\ &= W^0 + W^1 + \cdots + W^{N-1} = \begin{cases} 0 & \text{if } j \neq k \\ N & \text{if } j = k. \end{cases} \end{aligned}$$

Indeed, when $j = k$, then $\bar{w}^k w^k = (\bar{w}w)^k = (e^{2\pi i/N} e^{-2\pi i/N})^k = 1^k = 1$, so that the sum of *these* N terms equals N ; these are the diagonal entries of $\mathbf{G}_N$. Also, when $j \neq k$, then $W \neq 1$ and we have a geometric sum (whose value is given by (6) in Sec. 15.1 with $q = W$ and $n = N - 1$)

$$W^0 + W^1 + \cdots + W^{N-1} = \frac{1 - W^N}{1 - W} = 0$$

because $W^N = (\bar{w}^j w^k)^N = (e^{2\pi i j} e^{-2\pi i k})^N = 1^j \cdot 1^k = 1$. ■

We have seen that $\hat{\mathbf{f}}$ is the frequency spectrum of the signal $f(x)$. Thus the components $\hat{f}_n$ of $\hat{\mathbf{f}}$ give a resolution of the 2π -periodic function $f(x)$ into simple (complex) harmonics. Here one should use only n 's that are much smaller than $N/2$, to avoid **aliasing**. By this we mean the effect caused by sampling at too few (equally spaced) points, so that, for instance, in a motion picture, rotating wheels appear as rotating too slowly or even in the wrong sense. Hence in applications, N is usually large. But this poses a problem. Eq. (18) requires $O(N)$ operations for any particular n , hence $O(N^2)$ operations for, say, all $n < N/2$. Thus, already for 1000 sample points the straightforward calculation would involve millions of operations. However, this difficulty can be overcome by the so-called **fast Fourier transform (FFT)**, for which codes are readily available (e.g., in Maple). The FFT is a computational method for the DFT that needs only $O(N) \log_2 N$ operations instead of $O(N^2)$. It makes the DFT a practical tool for large N . Here one chooses $N = 2^p$ (p integer) and uses the special form of the Fourier matrix to break down the given problem into smaller problems. For instance, when $N = 1000$, those operations are reduced by a factor $1000/\log_2 1000 \approx 100$.

The breakdown produces two problems of size $M = N/2$. This breakdown is possible because for $N = 2M$ we have in (19)

$$w_N^2 = w_{2M}^2 = (e^{-2\pi i/N})^2 = e^{-4\pi i/(2M)} = e^{-2\pi i/M} = w_M.$$

The given vector $\mathbf{f} = [f_0 \cdots f_{N-1}]^T$ is split into two vectors with M components each, namely, $\mathbf{f}_{\text{ev}} = [f_0 \ f_2 \ \cdots \ f_{N-2}]^T$ containing the even components of $\mathbf{f}$, and $\mathbf{f}_{\text{od}} = [f_1 \ f_3 \ \cdots \ f_{N-1}]^T$ containing the odd components of $\mathbf{f}$. For $\mathbf{f}_{\text{ev}}$ and $\mathbf{f}_{\text{od}}$ we determine the DFTs

$$\hat{\mathbf{f}}_{\text{ev}} = [\hat{f}_{\text{ev},0} \ \hat{f}_{\text{ev},2} \ \cdots \ \hat{f}_{\text{ev},N-2}]^T = \mathbf{F}_M \mathbf{f}_{\text{ev}}$$

and

$$\hat{\mathbf{f}}_{\text{od}} = [\hat{f}_{\text{od},1} \ \hat{f}_{\text{od},3} \ \cdots \ \hat{f}_{\text{od},N-1}]^T = \mathbf{F}_M \mathbf{f}_{\text{od}}$$

involving the same $M \times M$ matrix $\mathbf{F}_M$. From these vectors we obtain the components of the DFT of the given vector f by the formulas

$$(22) \quad \begin{aligned} (a) \quad \hat{f}_n &= \hat{f}_{\text{ev},n} + w_N^n \hat{f}_{\text{od},n} & n = 0, \dots, M-1 \\ (b) \quad \hat{f}_{n+M} &= \hat{f}_{\text{ev},n} - w_N^n \hat{f}_{\text{od},n} & n = 0, \dots, M-1. \end{aligned}$$

For $N = 2^p$ this breakdown can be repeated $p - 1$ times in order to finally arrive at $N/2$ problems of size 2 each, so that the number of multiplications is reduced as indicated above.

We show the reduction from $N = 4$ to $M = N/2 = 2$ and then prove (22).

EXAMPLE 5 Fast Fourier Transform (FFT). Sample of $N = 4$ Values

When $N = 4$, then $w = w_N = -i$ as in Example 4 and $M = N/2 = 2$, hence $w = w_M = e^{-2\pi i/2} = e^{-\pi i} = -1$. Consequently,

$$\begin{aligned}\hat{\mathbf{f}}_{\text{ev}} &= \begin{bmatrix} \hat{f}_0 \\ \hat{f}_2 \end{bmatrix} = \mathbf{F}_2 \mathbf{f}_{\text{ev}} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} f_0 \\ f_2 \end{bmatrix} = \begin{bmatrix} f_0 + f_2 \\ f_0 - f_2 \end{bmatrix} \\ \hat{\mathbf{f}}_{\text{od}} &= \begin{bmatrix} \hat{f}_1 \\ \hat{f}_3 \end{bmatrix} = \mathbf{F}_2 \mathbf{f}_{\text{od}} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} f_1 \\ f_3 \end{bmatrix} = \begin{bmatrix} f_1 + f_3 \\ f_1 - f_3 \end{bmatrix}.\end{aligned}$$

From this and (22a) we obtain

$$\begin{aligned}\hat{f}_0 &= \hat{f}_{\text{ev},0} + w_N^0 \hat{f}_{\text{od},0} = (f_0 + f_2) + (f_1 + f_3) = f_0 + f_1 + f_2 + f_3 \\ \hat{f}_1 &= \hat{f}_{\text{ev},1} + w_N^1 \hat{f}_{\text{od},1} = (f_0 - f_2) - i(f_1 + f_3) = f_0 - if_1 - f_2 + if_3.\end{aligned}$$

Similarly, by (22b),

$$\begin{aligned}\hat{f}_2 &= \hat{f}_{\text{ev},0} - w_N^0 \hat{f}_{\text{od},0} = (f_0 + f_2) - (f_1 + f_3) = f_0 - f_1 + f_2 - f_3 \\ \hat{f}_3 &= \hat{f}_{\text{ev},1} - w_N^1 \hat{f}_{\text{od},1} = (f_0 - f_2) - (-i)(f_1 + f_3) = f_0 + if_1 - f_2 - if_3.\end{aligned}$$

This agrees with Example 4, as can be seen by replacing 0, 1, 4, 9 with f_0, f_1, f_2, f_3 . ■

We prove (22). From (18) and (19) we have for the components of the DFT

$$\hat{f}_n = \sum_{k=0}^{N-1} w_N^{kn} f_k.$$

Splitting into two sums of $M = N/2$ terms each gives

$$\hat{f}_n = \sum_{k=0}^{M-1} w_N^{2kn} f_{2k} + \sum_{k=0}^{M-1} w_N^{(2k+1)n} f_{2k+1}.$$

We now use $w_N^2 = w_M$ and pull out w_N^n from under the second sum, obtaining

$$(23) \quad \hat{f}_n = \sum_{k=0}^{M-1} w_M^{kn} f_{\text{ev},k} + w_N^n \sum_{k=0}^{M-1} w_M^{kn} f_{\text{od},k}.$$

The two sums are $f_{\text{ev},n}$ and $f_{\text{od},n}$, the components of the “half-size” transforms $\mathbf{Ff}_{\text{ev}}$ and $\mathbf{Ff}_{\text{od}}$.

Formula (22a) is the same as (23). In (22b) we have $n + M$ instead of n . This causes a sign change in (23), namely $-w_N^n$ before the second sum because

$$w_N^M = e^{-2\pi i M/N} = e^{-2\pi i/2} = e^{-\pi i} = -1.$$

This gives the minus in (22b) and completes the proof. ■

PROBLEM SET 11.9

- 1. Review in complex.** Show that $1/i = -i$, $e^{-ix} = \cos x - i \sin x$, $e^{ix} + e^{-ix} = 2 \cos x$, $e^{ix} - e^{-ix} = 2i \sin x$, $e^{ikx} = \cos kx + i \sin kx$.

2–11 FOURIER TRANSFORMS BY INTEGRATION

Find the Fourier transform of $f(x)$ (without using Table III in Sec. 11.10). Show details.

2. $f(x) = \begin{cases} e^{2ix} & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$
3. $f(x) = \begin{cases} 1 & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$
4. $f(x) = \begin{cases} e^{kx} & \text{if } x < 0 \quad (k > 0) \\ 0 & \text{if } x > 0 \end{cases}$
5. $f(x) = \begin{cases} e^x & \text{if } -a < x < a \\ 0 & \text{otherwise} \end{cases}$
6. $f(x) = e^{-|x|} \quad (-\infty < x < \infty)$
7. $f(x) = \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$
8. $f(x) = \begin{cases} xe^{-x} & \text{if } -1 < x < 0 \\ 0 & \text{otherwise} \end{cases}$
9. $f(x) = \begin{cases} |x| & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$
10. $f(x) = \begin{cases} x & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$
11. $f(x) = \begin{cases} -1 & \text{if } -1 < x < 0 \\ 1 & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

12–17 USE OF TABLE III IN SEC. 11.10. OTHER METHODS

12. Find $\mathcal{F}(f(x))$ for $f(x) = xe^{-x}$ if $x > 0$, $f(x) = 0$ if $x < 0$, by (9) in the text and formula 5 in Table III (with $a = 1$). *Hint.* Consider xe^{-x} and e^{-x} .
13. Obtain $\mathcal{F}(e^{-x^2/2})$ from Table III.
14. In Table III obtain formula 7 from formula 8.
15. In Table III obtain formula 1 from formula 2.
16. **TEAM PROJECT. Shifting (a)** Show that if $f(x)$ has a Fourier transform, so does $f(x - a)$, and $\mathcal{F}\{f(x - a)\} = e^{-iwa}\mathcal{F}\{f(x)\}$.
(b) Using (a), obtain formula 1 in Table III, Sec. 11.10, from formula 2.
(c) Shifting on the w -Axis. Show that if $\hat{f}(w)$ is the Fourier transform of $f(x)$, then $\hat{f}(w - a)$ is the Fourier transform of $e^{iax}f(x)$.
(d) Using (c), obtain formula 7 in Table III from 1 and formula 8 from 2.
17. What could give you the idea to solve Prob. 11 by using the solution of Prob. 9 and formula (9) in the text? Would this work?

18–25 DISCRETE FOURIER TRANSFORM

18. Verify the calculations in Example 4 of the text.
19. Find the transform of a general signal $f = [f_1 \ f_2 \ f_3 \ f_4]^T$ of four values.
20. Find the inverse matrix in Example 4 of the text and use it to recover the given signal.
21. Find the transform (the frequency spectrum) of a general signal of two values $[f_1 \ f_2]^T$.
22. Recreate the given signal in Prob. 21 from the frequency spectrum obtained.
23. Show that for a signal of eight sample values, $w = e^{-i/4} = (1 - i)/\sqrt{2}$. Check by squaring.
24. Write the Fourier matrix $\mathbf{F}$ for a sample of eight values explicitly.
25. **CAS Problem.** Calculate the inverse of the 8×8 Fourier matrix. Transform a general sample of eight values and transform it back to the given data.

11.10 Tables of Transforms

Table I. Fourier Cosine Transforms

See (2) in Sec. 11.8.

	$f(x)$	$\hat{f}_c(w) = \mathcal{F}_c(f)$	
1	$\begin{cases} 1 & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$	$\sqrt{\frac{2}{\pi}} \frac{\sin aw}{w}$	
2	$x^{a-1} \quad (0 < a < 1)$	$\sqrt{\frac{2}{\pi}} \frac{\Gamma(a)}{w^a} \cos \frac{a\pi}{2}$	$(\Gamma(a) \text{ see App. A3.1.})$
3	$e^{-ax} \quad (a > 0)$	$\sqrt{\frac{2}{\pi}} \left(\frac{a}{a^2 + w^2} \right)$	
4	$e^{-x^2/2}$	$e^{-w^2/2}$	
5	$e^{-ax^2} \quad (a > 0)$	$\frac{1}{\sqrt{2a}} e^{-w^2/(4a)}$	
6	$x^n e^{-ax} \quad (a > 0)$	$\sqrt{\frac{2}{\pi}} \frac{n!}{(a^2 + w^2)^{n+1}} \operatorname{Re}(a + iw)^{n+1}$	$\operatorname{Re} =$ Real part
7	$\begin{cases} \cos x & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$	$\frac{1}{\sqrt{2\pi}} \left[\frac{\sin a(1-w)}{1-w} + \frac{\sin a(1+w)}{1+w} \right]$	
8	$\cos(ax^2) \quad (a > 0)$	$\frac{1}{\sqrt{2a}} \cos\left(\frac{w^2}{4a} - \frac{\pi}{4}\right)$	
9	$\sin(ax^2) \quad (a > 0)$	$\frac{1}{\sqrt{2a}} \cos\left(\frac{w^2}{4a} + \frac{\pi}{4}\right)$	
10	$\frac{\sin ax}{x} \quad (a > 0)$	$\sqrt{\frac{\pi}{2}} (1 - u(w-a))$	(See Sec. 6.3.)
11	$\frac{e^{-x} \sin x}{x}$	$\frac{1}{\sqrt{2\pi}} \arctan \frac{2}{w^2}$	
12	$J_0(ax) \quad (a > 0)$	$\sqrt{\frac{2}{\pi}} \frac{1}{\sqrt{a^2 - w^2}} (1 - u(w-a))$	$(\text{See Secs. 5.5, 6.3.})$

Table II. Fourier Sine Transforms

See (5) in Sec. 11.8.

	$f(x)$	$\hat{f}_s(w) = \mathcal{F}_s(f)$
1	$\begin{cases} 1 & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$	$\sqrt{\frac{2}{\pi}} \left[\frac{1 - \cos aw}{w} \right]$
2	$1/\sqrt{x}$	$1/\sqrt{w}$
3	$1/x^{3/2}$	$2\sqrt{w}$
4	$x^{a-1} \quad (0 < a < 1)$	$\sqrt{\frac{2}{\pi}} \frac{\Gamma(a)}{w^a} \sin \frac{a\pi}{2} \quad (\Gamma(a) \text{ see App. A3.1.})$
5	$e^{-ax} \quad (a > 0)$	$\sqrt{\frac{2}{\pi}} \left(\frac{w}{a^2 + w^2} \right)$
6	$\frac{e^{-ax}}{x} \quad (a > 0)$	$\sqrt{\frac{2}{\pi}} \arctan \frac{w}{a}$
7	$x^n e^{-ax} \quad (a > 0)$	$\sqrt{\frac{2}{\pi}} \frac{n!}{(a^2 + w^2)^{n+1}} \operatorname{Im}(a + iw)^{n+1} \quad \begin{matrix} \operatorname{Im} = \\ \text{Imaginary part} \end{matrix}$
8	$xe^{-x^2/2}$	$w e^{-w^2/2}$
9	$xe^{-ax^2} \quad (a > 0)$	$\frac{w}{(2a)^{3/2}} e^{-w^2/4a}$
10	$\begin{cases} \sin x & \text{if } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$	$\frac{1}{\sqrt{2\pi}} \left[\frac{\sin a(1-w)}{1-w} - \frac{\sin a(1+w)}{1+w} \right]$
11	$\frac{\cos ax}{x} \quad (a > 0)$	$\sqrt{\frac{\pi}{2}} u(w-a) \quad (\text{See Sec. 6.3.})$
12	$\arctan \frac{2a}{x} \quad (a > 0)$	$\sqrt{2\pi} \frac{\sin aw}{w} e^{-aw}$

Table III. Fourier Transforms

See (6) in Sec. 11.9.

	$f(x)$	$\hat{f}(w) = \mathcal{F}(f)$
1	$\begin{cases} 1 & \text{if } -b < x < b \\ 0 & \text{otherwise} \end{cases}$	$\sqrt{\frac{2}{\pi}} \frac{\sin bw}{w}$
2	$\begin{cases} 1 & \text{if } b < x < c \\ 0 & \text{otherwise} \end{cases}$	$\frac{e^{-ibw} - e^{-icw}}{iw\sqrt{2\pi}}$
3	$\frac{1}{x^2 + a^2} \quad (a > 0)$	$\sqrt{\frac{\pi}{2}} \frac{e^{-a w }}{a}$
4	$\begin{cases} x & \text{if } 0 < x < b \\ 2x - b & \text{if } b < x < 2b \\ 0 & \text{otherwise} \end{cases}$	$\frac{-1 + 2e^{ibw} - e^{2ibw}}{\sqrt{2\pi}w^2}$
5	$\begin{cases} e^{-ax} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases} \quad (a > 0)$	$\frac{1}{\sqrt{2\pi}(a + iw)}$
6	$\begin{cases} e^{ax} & \text{if } b < x < c \\ 0 & \text{otherwise} \end{cases}$	$\frac{e^{(a-iw)c} - e^{(a-iw)b}}{\sqrt{2\pi}(a - iw)}$
7	$\begin{cases} e^{iax} & \text{if } -b < x < b \\ 0 & \text{otherwise} \end{cases}$	$\sqrt{\frac{2}{\pi}} \frac{\sin b(w - a)}{w - a}$
8	$\begin{cases} e^{iax} & \text{if } b < x < c \\ 0 & \text{otherwise} \end{cases}$	$\frac{i}{\sqrt{2\pi}} \frac{e^{ib(a-w)} - e^{ic(a-w)}}{a - w}$
9	$e^{-ax^2} \quad (a > 0)$	$\frac{1}{\sqrt{2a}} e^{-w^2/4a}$
10	$\frac{\sin ax}{x} \quad (a > 0)$	$\sqrt{\frac{\pi}{2}} \quad \text{if } w < a; \quad 0 \text{ if } w > a$

CHAPTER 11 REVIEW QUESTIONS AND PROBLEMS

1. What is a Fourier series? A Fourier cosine series? A half-range expansion? Answer from memory.
 2. What are the Euler formulas? By what very important idea did we obtain them?
 3. How did we proceed from 2π -periodic to general-periodic functions?
 4. Can a discontinuous function have a Fourier series? A Taylor series? Why are such functions of interest to the engineer?
 5. What do you know about convergence of a Fourier series? About the Gibbs phenomenon?
 6. The output of an ODE can oscillate several times as fast as the input. How come?
 7. What is approximation by trigonometric polynomials? What is the minimum square error?
 8. What is a Fourier integral? A Fourier sine integral? Give simple examples.
 9. What is the Fourier transform? The discrete Fourier transform?
 10. What are Sturm–Liouville problems? By what idea are they related to Fourier series?
- 11–20 FOURIER SERIES.** In Probs. 11, 13, 16, 20 find the Fourier series of $f(x)$ as given over one period and sketch $f(x)$ and partial sums. In Probs. 12, 14, 15, 17–19 give answers, with reasons. Show your work detail.
11. $f(x) = \begin{cases} 0 & \text{if } -2 < x < 0 \\ 2 & \text{if } 0 < x < 2 \end{cases}$
 12. Why does the series in Prob. 11 have no cosine terms?
 13. $f(x) = \begin{cases} 0 & \text{if } -1 < x < 0 \\ x & \text{if } 0 < x < 1 \end{cases}$
 14. What function does the series of the cosine terms in Prob. 13 represent? The series of the sine terms?
 15. What function do the series of the cosine terms and the series of the sine terms in the Fourier series of e^x ($-5 < x < 5$) represent?
 16. $f(x) = |x|$ ($-\pi < x < \pi$)
 17. Find a Fourier series from which you can conclude that $1 - 1/3 + 1/5 - 1/7 + \cdots = \pi/4$.
 18. What function and series do you obtain in Prob. 16 by (termwise) differentiation?
 19. Find the half-range expansions of $f(x) = x$ ($0 < x < 1$).
 20. $f(x) = 3x^2$ ($-\pi < x < \pi$)
- 21–22 GENERAL SOLUTION**
- Solve, $y'' + \omega^2 y = r(t)$, where $|\omega| \neq 0, 1, 2, \dots$, $r(t)$ is 2π -periodic and
21. $r(t) = 3t^2$ ($-\pi < t < \pi$)
 22. $r(t) = |t|$ ($-\pi < t < \pi$)
- 23–25 MINIMUM SQUARE ERROR**
23. Compute the minimum square error for $f(x) = x/\pi$ ($-\pi < x < \pi$) and trigonometric polynomials of degree $N = 1, \dots, 5$.
 24. How does the minimum square error change if you multiply $f(x)$ by a constant k ?
 25. Same task as in Prob. 23, for $f(x) = |x|/\pi$ ($-\pi < x < \pi$). Why is E^* now much smaller (by a factor 100, approximately!)?
- 26–30 FOURIER INTEGRALS AND TRANSFORMS**
- Sketch the given function and represent it as indicated. If you have a CAS, graph approximate curves obtained by replacing ∞ with finite limits; also look for Gibbs phenomena.
26. $f(x) = x + 1$ if $0 < x < 1$ and 0 otherwise; by the Fourier sine transform
 27. $f(x) = x$ if $0 < x < 1$ and 0 otherwise; by the Fourier integral
 28. $f(x) = kx$ if $a < x < b$ and 0 otherwise; by the Fourier transform
 29. $f(x) = x$ if $1 < x < a$ and 0 otherwise; by the Fourier cosine transform
 30. $f(x) = e^{-2x}$ if $x > 0$ and 0 otherwise; by the Fourier transform

SUMMARY OF CHAPTER 11

Fourier Analysis. Partial Differential Equations (PDEs)

Fourier series concern **periodic functions** $f(x)$ of period $p = 2L$, that is, by definition $f(x + p) = f(x)$ for all x and some fixed $p > 0$; thus, $f(x + np) = f(x)$ for any integer n . These series are of the form

$$(1) \quad f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{L}x + b_n \sin \frac{n\pi}{L}x \right) \quad (\text{Sec. 11.2})$$

with coefficients, called the **Fourier coefficients** of $f(x)$, given by the Euler formulas (Sec. 11.2)

$$(2) \quad \begin{aligned} a_0 &= \frac{1}{2L} \int_{-L}^L f(x) dx, & a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx \\ b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx \end{aligned}$$

where $n = 1, 2, \dots$. For period 2π we simply have (Sec. 11.1)

$$(1^*) \quad f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

with the *Fourier coefficients* of $f(x)$ (Sec. 11.1)

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx.$$

Fourier series are fundamental in connection with periodic phenomena, particularly in models involving differential equations (Sec. 11.3, Chap. 12). If $f(x)$ is even [$f(-x) = f(x)$] or odd [$f(-x) = -f(x)$], they reduce to **Fourier cosine** or **Fourier sine series**, respectively (Sec. 11.2). If $f(x)$ is given for $0 \leq x \leq L$ only, it has two **half-range expansions** of period $2L$, namely, a cosine and a sine series (Sec. 11.2).

The set of cosine and sine functions in (1) is called the **trigonometric system**. Its most basic property is its **orthogonality** on an interval of length $2L$; that is, for all integers m and $n \neq m$ we have

$$\int_{-L}^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = 0, \quad \int_{-L}^L \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx = 0$$

and for all integers m and n ,

$$\int_{-L}^L \cos \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx = 0.$$

This orthogonality was crucial in deriving the Euler formulas (2).

Partial sums of Fourier series minimize the **square error** (Sec. 11.4).

Replacing the trigonometric system in (1) by other orthogonal systems first leads to **Sturm–Liouville problems** (Sec. 11.5), which are boundary value problems for ODEs. These problems are **eigenvalue problems** and as such involve a parameter λ that is often related to frequencies and energies. The solutions to Sturm–Liouville problems are called **eigenfunctions**. Similar considerations lead to other orthogonal series such as **Fourier–Legendre series** and **Fourier–Bessel series** classified as **generalized Fourier series** (Sec. 11.6).

Ideas and techniques of Fourier series extend to nonperiodic functions $f(x)$ defined on the entire real line; this leads to the **Fourier integral**

$$(3) \quad f(x) = \int_0^{\infty} [A(w) \cos wx + B(w) \sin wx] dw \quad (\text{Sec. 11.7})$$

where

$$(4) \quad A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos wv \, dv, \quad B(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \sin wv \, dv$$

or, in complex form (Sec. 11.9),

$$(5) \quad f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(w) e^{iwx} dw \quad (i = \sqrt{-1})$$

where

$$(6) \quad \hat{f}(w) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} dx.$$

Formula (6) transforms $f(x)$ into its **Fourier transform** $\hat{f}(w)$, and (5) is the inverse transform.

Related to this are the **Fourier cosine transform** (Sec. 11.8)

$$(7) \quad \hat{f}_c(w) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos wx \, dx$$

and the **Fourier sine transform** (Sec. 11.8)

$$(8) \quad \hat{f}_s(w) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin wx \, dx.$$

The **discrete Fourier transform (DFT)** and a practical method of computing it, called the **fast Fourier transform (FFT)**, are discussed in Sec. 11.9.